

From foreign demand to domestic production: A multi-country framework

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April 22, 2026

Abstract

This paper proposes a novel empirical strategy to study how third-country demand effects determine domestic production. I show that multilateral market access is a key determinant of a country's output composition in a multi-country trade model with non-homothetic CES preferences and economies of scale. The model predicts that firms' production choice is determined by a trade-off between demand incentives and production costs, and is reflected in the destination composition of a country's exports. Using product-level trade data, I confirm that market access increases exports disproportionately to high-income countries - suggesting a higher share of high-quality

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[†]This paper benefited from funding by the Leibniz Junior Research Group Grant J212/2025. This paper was previously circulated under the title "The geography of income and export patterns" and won the Young Economist Award at the 16th FIW Conference in International Economics. Thanks for valuable comments to Jonathan Dingel, Dave Donaldson, Carsten Eckel, Lisandra Flach, Rafael Greminger, Elsa Leromain, Florian Mayneris, Dennis Novy, Guzman Ourens, Sjak Smulders, Alan Spearot, Claudia Steinwender, Catherine Thomas, Gonzague Vannoorenberghe, and participants of the 24. and 27. Göttinger Workshop für Internationale Wirtschaftsbeziehungen, Antwerp University Economics Seminar, LMU/ifo Trade Winter Retreat 2023, 16th FIW Conference International Economics, CRC Retreat 2024 Tutzing, University Mainz Economics Seminar, the LSE-CEP Trade & Urban workshop, and the CEPR WEARE seminar series.

producers - yet only for exporters in the top quintile of the market access distribution.

Key words: Non-homothetic preferences, home market effect, quality differentiation

JEL Codes: F14, R12, O19

1 Introduction

Countries tend to export more of goods for which they have a relatively large domestic market. This insight, known as the ‘home market effect’ (Krugman, 1980; Fajgelbaum et al., 2011; Costinot et al., 2019), has been foundational for understanding how demand shapes the location of production. Home market effects lend support to the idea that import protection serves to promote exports (Costinot et al., 2019). An alternative policy implication arises from third-country demand effects (Behrens et al., 2009; Matsuyama, 2017; Lugovskyy and Skiba, 2015; Dingel, 2017). Access to demand in neighboring countries, or countries with which trade agreements are in place, allows firms to reap economies of scale, too. In that case, market integration acts as a *de facto* tool for industrial policy: trade policy affects the demand composition firms are exposed to, thereby changing their production incentives across locations.

This paper develops a novel empirical strategy to study the effect of third-country demand on domestic production. The strategy is grounded in a theory of international trade in vertically differentiated goods.¹ The idea is that, in bilateral trade, supply and demand act as complements so that a country’s export destinations reflect the variety of goods its firms are producing. The variety of goods firms are producing in a particular location, in turn, depends on their multilateral market access. Using product-level bilateral trade data, I show that multilateral market access increases exports differentially across destinations only for exporters in the top quintile of the market potential distribution. This is consistent with a trade-off in firms’ entry decision: quality upgrading pays off only if third-country demand compensates for differentiation costs.

Three challenges complicate the empirical study. First, per capita real income, the observational demand shifter for quality (Linder, 1961; Hallak, 2010; Fajgelbaum et al., 2011; Lugovskyy and Skiba, 2015; Dingel, 2017), is tied to the availability of high-quality inputs such as high-skilled labor (Verhoogen, 2008; Kugler and Verhoogen, 2012; Manova and Zhang, 2012; Caron et al., 2014; Fieler et al., 2018; Caron et al.,

¹For instance, consider different quality levels of apparel or toys. Demand effects are particularly relevant in the context of product quality differentiation. Better product quality is costly to produce and firms need to charge a quality premium to recover costs. Firms therefore benefit from proximity to consumers willing to pay such a quality premium. A different application with strong demand effects in production is pharmaceuticals that target locally concentrated diseases (Costinot et al., 2019).

2020). It is thus unclear whether richer countries trade more with each other in the spirit of Linder (1961) because of demand- or supply-side forces. Second, product quality is not directly observable.² Third, the firm distribution across quality levels for many countries with different global market access is not observable either.

To address these challenges, I develop a stylized model of international trade in which consumers have non-homothetic preferences and firms produce vertically differentiated varieties with increasing returns to scale. I derive the model’s prediction about the within-product firm distribution across quality levels and its implications for bilateral product-level trade. I then show how to isolate third-country demand effects from supply-side drivers of production in a structural gravity equation, similar to Costinot et al.’s ‘weak’ home market effect test (Costinot et al., 2019). I finally test the model prediction with publicly available trade data.

The key innovation of my theory is to extend the ‘home market effect’-logic by maintaining a role for third-country demand effects. I assume a many country-world with heterogeneous bilateral trade costs. Firms produce a single, vertically differentiated variety. Varieties differ in their quality, and firms choose which variety to supply by balancing fixed costs of production with global revenues. Global revenues vary across (i) quality segments and (ii) locations because of differences in access to consumers of different incomes. Consumers have non-homothetic CES preferences à la Comin et al. (2021) such that richer households spend larger shares of their income on high-quality varieties. Firm entry across the quality distribution is determined by a trade-off between catering to nearby-consumers preferences, the demand channel, and avoiding too high costs and too strong competition. Better market access to high-income countries globally allows more firms in a country to produce high-quality varieties and still break even.

In the aggregate, the firm distribution across quality segments determines bilateral trade patterns. This result provides a strategy to overcome the lack of data on product quality since bilateral trade data is readily available. The model implies a

²Several studies resort to quality measures for specific goods like wine (Crozet et al., 2012) or washing machines (Jaimovich et al., 2023). Studies using firm-level data often interpret unit values as proxies for quality (Manova and Yu (2017); Dingel (2017)). Other studies follow Khandelwal (2010) to infer product quality from a demand residual.

non-homothetic gravity equation at the product-level. Aside from standard determinants, bilateral trade depends on the complementarity between quality supply and demand: A country exports more to relatively rich destinations, with higher demand for high-quality varieties, if more firms supply high-end varieties. Thus, where a country exports to is informative about what its firms produce.³ Yet, what local firms produce depends on the multilateral market access the country has. My empirical strategy is therefore to test whether countries with better multilateral access to rich consumers export bilaterally more as destination income grows. Multilateral market access - or “foreign market potential” - is measured by a trade cost-weighted average of foreign countries’ per capita income. Trade costs are measured in geographic, cultural and monetary terms with the standard proxy variables.

I leverage three advantages of my model to disentangle demand- from supply-side drivers of trade patterns. First, because income and prices affect expenditure separately under non-homothetic CES preferences, I can control for a host of supply-side determinants through exporter-product fixed effects. Second, by focussing on third-country demand, I can control for domestic per capita income without taking a stance on whether this reflects domestic demand for quality or access to high-skilled labor. Third, I rely on a threshold condition: Firms sort into high-end production only if the incentive from proximity to high-income consumers to do so is strong enough to overcome the disincentive from incurring higher unit costs. In line with the model’s prediction, I find that exports increase with destination income only for countries with high “market potential”, the multilateral measure of integration with high-income countries.

The empirical results are consistent with the theory’s predictions. I estimate that, all else equal, only countries in the top quintile of the market potential distribution export disproportionately to high-income countries. This aligns with the idea that third-country demand affects a country’s output quality but only if they are strong enough to outweigh differentiation costs. Additionally, I provide evidence that the

³The logic applied here is reminiscent of a revealed preferences approach and microfoundations the Linder hypothesis (Linder, 1961). See also the discussion and empirical analysis of the Linder hypothesis in Hallak (2006, 2010).

effect is stronger when exporter-product unit values differ less across destinations - supportive of the idea of destination pooling by single-variety producers. It is worth noting that while the assumption of single variety producers simplifies the theory, it is not crucial for third-country demand effects to matter. I show how to extend the model to allow for product customization across destinations, in line with evidence from several country-specific, firm-level studies⁴, and discuss implications for the estimation strategy in section 3.

Related literature The paper makes contributions to several strands of the literature. First, the paper provides new theoretical and empirical evidence for the role of location for specialization. It adds to a large number of studies on the home market effect more generally, among others Krugman (1980); Hanson and Xiang (2004); Erhardt (2017); Costinot et al. (2019). Closely related studies are especially Behrens et al. (2009), Matsuyama (2017), and Norris (2023), which discuss the conditions of a home market effect in a many-country-world for homogeneous goods. Relative to these papers, my model and estimation strategy yields a novel insight: namely, that demand effects not only raise the level of exports but change the destination composition of exports. This paper relates furthermore in spirit to Morales et al. (2019); Forge et al. (2024) and Fajgelbaum et al. (2024) which also study third-country effects in trade.

Second, the paper improves our understanding of the role of demand for the supply of product quality. The paper thereby also contributes to the continuously growing literature on trade with non-homothetic preferences, e.g. Hallak (2006, 2010); Fieler (2011); Baldwin and Harrigan (2011); Markusen (2013); Caron et al. (2014); Fajgelbaum and Khandelwal (2016); Comin et al. (2021); Hillrichs and Vannoorenberghe (2022, 2024). The paper builds on previous studies on the role of demand for quality supply, notably Fajgelbaum et al. (2011); Dingel (2017) and Lugovskyy and Skiba (2015).⁵ What sets the present paper apart from the previous studies is the assumption of non-homothetic CES preferences, introduced by Comin et al. (2021).⁶ The

⁴See, among others, Bastos and Silva (2010), Manova and Zhang (2012), and Flach (2016) on quality market segmentation using firm-level data for Portugal, China, and Brazil, respectively.

⁵The role of domestic market integration for product quality is also analysed in the context of India by Kong (2025).

⁶Fajgelbaum et al. (2011) and Dingel (2017) assume a discrete demand system with variation in quality

main advantage of working with non-homothetic CES preferences is that income and prices affect expenditure separately in the implied demand system. This allows me to isolate the demand mechanism from supply-side determinants of product quality in the empirical analysis. Compared to Fajgelbaum et al. (2011) and Dingel (2017), the paper also makes progress by deriving theoretical results under heterogeneous trade costs. Empirically, the paper differs from Dingel (2017) and Lugovskyy and Skiba (2015) in the data employed and, therefore, the estimation strategy. In the present paper, I infer product quality based on the export shares across destinations of varying incomes. This indirect strategy, with a clear structural foundation, is appropriate for aggregate data. In contrast, Dingel (2017) and Lugovskyy and Skiba (2015) infer product quality from unit values, which is more appropriate when micro-level information about confounding factors like productivity and mark-ups is available. The advantage of employing product- rather than firm-level data is the larger variation in destination income and market access to identify the role of demand in quality specialization. To connect to the previous publications, I analyse unit values in the final section of the paper.

The paper also complements the literature on supply-side determinants of quality differentiation. Several studies show that firms that sell better quality also use higher quality inputs (Kugler and Verhoogen (2012); Manova and Zhang (2012); Manova and Yu (2017)). From a policy perspective, while these papers call for import liberalization to promote investments in product quality, my paper adds nuance by considering the trade-off between costs of producing high-quality goods and having access to demand for high-quality goods. Imports of high-quality inputs should be more strongly promoted in locations remote from demand and accompanied by means to ease access to demand.

Finally, the paper adds to the literature on trade and development. The paper's results suggest that integration with rich economies helps developing countries avoid

across goods rather than within goods. Similarly, in Lugovskyy and Skiba (2015), bilateral trade flows are a constant fraction of income and do not exhibit the non-homothetic gravity structure. Liu and Meissner (2019) choose AIDS preferences to derive a “non-homothetic market potential”. However, gravity estimates of an AIDS framework are highly sensitive to outliers as shown in Hillrichs and Vannoorenberghe (2022). Therefore, I prefer a CES setup.

the “home market trap” in quality supply by shifting producers’ incentives towards high-quality production. In spirit related is the experimental study by Atkin et al. (2017) which shows that access to high-income markets positively affects Egyptian rug producers’ productivity and output quality (see also Bold et al. (2022) for experimental evidence in the agricultural sector). Goldberg and Reed (2023) highlight that too little domestic demand can pose a barrier to development, which international integration can help overcome. The authors point out that integration notably with richer countries provides a larger boost for development. Frazer and Van Biesebroeck (2010) show that developing countries with preferential market access to the US also sell more to Europe. These results suggest that trade liberalization with one high-income country may pave the way to trade with other high-income countries via the demand-for-quality channel.

The next section develops a conceptual framework in which the geography of income affects comparative advantage in product quality. Section 3 explains the empirical strategy to test for demand-driven comparative advantage and discusses identification and robustness to theoretical extensions. Section 4 describes the data and section 5 presents results. Section 6 concludes.

2 Conceptual Framework

I begin with a theoretical analysis that spells out the conditions for third-country-demand effects on the firm distribution across quality levels. To build intuition, I keep the supply-side highly stylized in the baseline model. In the appendix A, I derive results for a generalized version of the model with less restrictive supply-side assumptions. The theoretical results inform the empirical strategy to identify demand effects for the location of production.

2.1 Model

Households Assume that a representative consumer in market d has income I_d which they spend to achieve utility $\bar{U}_d$.⁷ The consumer derives utility from consuming a bundle of differentiated products $j = 1, \dots, J$. From each product, the consumer purchases vertically differentiated varieties $v = 1, \dots, V_j$ from a set of countries $o = 1, \dots, K$. Each country hosts firms producing horizontally differentiated versions ω of variety v . The total number of firms in country o , M_o , is exogenous while the distribution of firms across vertically differentiated varieties is endogenously determined. Consumer's utility in d is implicitly defined by

$$\bar{U}_d = C_d = C_{0d}^{\eta_{0d}} \prod_{j=1}^J C_{jd}^{\eta_{jd}}, \quad \text{where} \quad (1)$$

$$C_{jd} = \left[\sum_{v=1}^{V_j} \tilde{\lambda}_{vjd}^{1-\rho} C_{vjd}^\rho \right]^{\frac{1}{\rho}} \quad \text{and}$$

$$C_{vjd} = \left(\sum_{o=1}^K \int_{\omega_o \in \Omega_{ovj}} c_{vjd}(\omega_o)^\zeta d\omega_o \right)^{\frac{1}{\zeta}} \quad \text{and}$$

$$\tilde{\lambda}_{vjd} = \frac{\alpha_{vjd} \lambda_{vjd}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} \lambda_{v'jd}} \quad \text{with} \quad \lambda_{vjd} = \bar{U}_d^{\delta_{vj}}. \quad (2)$$

The substitution parameters $\rho = \frac{\sigma-1}{\sigma}$ and $\zeta = \frac{\xi-1}{\xi}$ are such that varieties are more substitutable than products, $\xi > \sigma > 1$. Ω_{ovj} is the set of all firms producing variety v in country o . $\alpha_{vjd} > 0$ is an exogenous, idiosyncratic taste shifter. The variable δ_{vj} governs the non-homotheticity as in Comin et al. (2021)⁸ and characterizes the vertically differentiated varieties v of good j . If $\delta_{vj} = \delta_j \forall v$, the utility takes the standard homothetic CES form.

The household's expenditure minimization problem yields optimal spending

$$x_{vjd}(\omega_o) \equiv p_{vjd}(\omega_o) c_{vjd}(\omega_o) = p_{vjd}(\omega_o)^{1-\xi} P_{vjd}^{\xi-\sigma} \alpha_{vjd} \lambda_{vjd} P_{jd}^{\sigma-1} \eta_{jd} I_d, \quad (3)$$

⁷I abstract here from within-country inequality for simplicity. The intuition developed in this section carries over to a model with within-country inequality.

⁸The expression for C_{jd} equals equation (1) in Comin et al. (2021) with the following definitions of the parameters in Comin et al. (2021) (replacing the index v with i): $\Upsilon_i = \frac{\alpha_i}{C_{jd}}$, $g(U) = \bar{U}_d$ and $\varepsilon_i = \frac{\delta_i}{\sigma}$.

where the price indices are defined as follows:

$$P_{vjd} = \left(\sum_{o=1}^K \int_{\omega_o \in \Omega_o} p_{vjd}(\omega_o)^{1-\xi} d\omega_o \right)^{\frac{1}{1-\xi}} \quad (4)$$

$$P_{jd} = \left(\sum_{v'=1}^{V_j} \alpha_{v'jd} \lambda_{v'jd} P_{v'jd}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

As shown in appendix A.1, utility and expenditure, and hence real income, are directly proportional so that from now I use $\lambda_{vjd} = I_d^{\delta_{vj}}$.

Firms Firms are located in countries $o = 1, \dots, K$. Firms in the *homogeneous* good sector $j = 0$ operate under constant returns to scale with productivity a_{o0} . Their product is freely traded. Labor is mobile between sectors but not across countries. Thus, the homogeneous good sector's productivity determines the wage rate in o and ensures trade is balanced.⁹

Firms in the *differentiated* goods sectors $j = 1, \dots, J$ operate under increasing returns to scale. Firms in sector j draw δ_{vj} from some distribution $\mathcal{D}_j$ and decide whether or not to produce variety v . To produce, firms incur a fixed cost $w_o f_j$ in terms of labor and have to employ $a_{oj} \kappa_{vj}$ workers to produce one unit of the differentiated good characterized by δ_{vj} . κ_{vj} is a generic, twice differentiable function of δ_{vj} , $\kappa_{vj} = f(\delta_{vj})$ with $\frac{\partial \kappa_{vj}}{\partial \delta_{vj}} > 0$. In a more general model setup presented in appendix A.1, I allow for differentiation costs to vary by exporter and destination. I keep the parameter constant across countries in the base model to focus attention on market access. To sell one unit of output to a destination d , firms need to ship τ_{od} units of the good, with $\tau_{oo} = 1, \tau_{od} > 1$ for $o \neq d$. Entry is free so that if profits are non-negative, firms decide to produce and maximize profits

$$\pi_{vj}(\omega_o) = \sum_{d=1}^K c_{vjd}(\omega_o) (p_{vjd}(\omega_o) - \tau_{od} \kappa_{vj} a_{oj} w_o) - w_o f_j \quad (5)$$

⁹See discussion in Crozet and Trionfetti (2008); Behrens et al. (2009); Lugovskyy and Skiba (2015). Other studies on market potential focus precisely on the role of location for wages, e.g. Redding and Venables (2004); Breinlich and Cuñat (2013); Liu and Meissner (2019). Those studies discuss the role of location for wages in the context of more aggregate sectors than this paper (e.g. Breinlich and Cunat study centrality with respect to manufacturing and agriculture.)

by charging the unit price

$$p_{vjd}(\omega_o) = \frac{\xi}{\xi - 1} \tau_{od} \kappa_{vj} a_{oj} w_o = p_{voj d}. \quad (6)$$

Equilibrium Given firms' pricing decision, all firms in the differentiated goods sector in country o make equal profits, $x_{vjd}(\omega_o) = x_{voj d} \forall \omega \in \Omega_o$. The price index of a variety vj in destination d , P_{vjd} , becomes

$$P_{vjd} = \frac{\xi}{\xi - 1} \kappa_{vj} \left[\sum_{o=1}^K (a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}} \quad (7)$$

where $M_{ovj} = \int_{\omega \in \Omega_o} d\omega$ is the number of active firms producing variety δ_{vj} in o and $\sum_{j=1}^J \sum_{v=1}^{V_j} M_{ovj} = M_o$. It will be useful to define country d 's *supplier access* for a variety v by

$$\widetilde{M}_{vjd} \equiv \sum_{o=1}^K \theta_{ojd} M_{ovj} \quad \text{where} \quad \theta_{ojd} \equiv \frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}}. \quad (8)$$

Substituting P_{vjd} from (7) in the expression for P_{jd} (4), using $\lambda_{vjd} = I_d^{\delta_{vj}}$, and rearranging yields

$$P_{jd}^{1-\sigma} k_{jd} = \sum_{v=1}^{V_j} \alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vj}^{1-\sigma} \widetilde{M}_{vjd}^{\frac{1-\sigma}{1-\xi}} \quad (9)$$

where $k_{jd} \equiv \left(\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi} \right)^{\frac{1-\sigma}{\xi-1}} \left(\frac{\xi}{\xi-1} \right)^{\sigma-1}$. Finally, substituting all price expressions in the demand equation (3), expenditure per firm equals

$$\begin{aligned} x_{voj d}(\omega) &= \frac{1}{M_{ovj}} s_{ovjd} s_{vjd} \eta_{jd} I_d \\ \text{where} \quad s_{ovjd} &= \left(\frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} M_{ovj} \right) / \widetilde{M}_{vjd} \\ \text{and} \quad s_{vjd} &= \frac{\alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vj}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} I_d^{\delta_{v'j}} \kappa_{v'j}^{1-\sigma} [\widetilde{M}_{v'jd}]^{\frac{1-\sigma}{1-\xi}}} \end{aligned} \quad (10)$$

in equilibrium. The expenditure share on variety v of good j sourced from country o , s_{ovjd} , increases if country o contributes relatively more to supplier access in d . Nearby-countries (low τ_{od}) with Ricardian comparative advantage in product j (low $a_{oj} w_o$)

and a large number of v -producers (M_{ovj}) lower the price index P_{vjd} more and hence command a higher share in d 's expenditure in equilibrium. The expenditure share on variety v of good j , s_{vjd} , is determined by a cost-competition-preference trade-off. Lower differentiation costs κ_{vj} and better supplier access $\widetilde{M}_{vjd}$ raise a variety's share in destination expenditure. With non-homothetic preferences, however, a higher income ($I_d^{\delta_{vj}}$) and a high taste draw α_{vjd} compensate a cost disadvantage of high- δ_{vj} varieties and those varieties that are predominantly produced in remote locations from the perspective of d .

The number of firms M_{ovj} per variety is pinned down in equilibrium by a zero profit condition. Global profits of a firm are equal to $\sum_{d=1}^K x_{vojd}/\xi - w_o f_j$. Because of free entry, firms will enter production of vj in o until profits are equal to zero for each firm. In equilibrium,

$$\Delta_{ovj} \equiv \sum_{d=1}^K (\tau_{od})^{1-\xi} \frac{s_{vjd} \eta_{jd} I_d}{\left[\sum_{o=1}^K (a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]} = \frac{\xi w_o f_j}{(a_{oj} w_o)^{1-\xi}}. \quad (11)$$

Δ_{ovj} measures origin o 's global market access for variety v . Given the constraint $\xi > \sigma > 1$, it follows that $\frac{\xi-\sigma}{1-\xi} < 0$ and Δ_{ovj} is downward sloping in M_{ovj} (see appendix A.2 for derivation). Given variable and fixed costs of production, a higher value of Δ_{ovj} increases entry into variety v . Δ_{ovj} maps worldwide market conditions for variety v to the country of origin through bilateral trade costs. Market conditions are described by per firm-expenditure on a variety v in a destination, adjusted for the productivity and trade costs of competitors in that market. Conditional on competition $\left[\sum_{o=1}^K (a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]$, proximity to markets with higher aggregate expenditure on variety v in equilibrium induces entry.

To connect to the previous literature on 'home market effects', it is useful to express the left-hand side of the zero profit condition as a composition of domestic and foreign demand conditions, and impose a symmetric geography such that trade costs are

uniform internationally, $\tau_{od} = \tau$ if $o \neq d$ and $\tau = 1$ if $o = d$:

$$\Delta_{ovj} = \underbrace{\frac{s_{vjo}\eta_{jo}I_o}{\left[\sum_{o=1}^K (a_{oj}w_o)^{1-\xi} M_{ovj}\right]}}_{\Delta_{ovj}^{home}} + \sum_{d \neq o}^K \underbrace{\frac{s_{vjd}\eta_{jd}I_d}{\tau^{1-\xi} \left[\sum_{o=1}^K (a_{oj}w_o)^{1-\xi} M_{ovj}\right]}}_{\Delta_{ovj}^{foreign}}.$$

When production costs are constant, $a_{oj}w_o = a_jw$, it is the domestic demand *only* that affects the selection of firms since $\Delta_{ovj}^{foreign} = \Delta_{vj}^{foreign} \forall o$.¹⁰

2.2 Analytical results

The model yields two important predictions. First, destination income affects exports differentially across countries when preferences are non-homothetic and products vertically differentiated. The income elasticity of demand for goods from a given origin is

$$\varepsilon_{ojd} \equiv \frac{dX_{ojd}}{dI_d} \frac{I_d}{X_{ojd}} = 1 + \sum_{v=1}^{V_j} \frac{s_{ovjd}s_{vjd}}{\sum_{v'=1}^{V_j} s_{ov'jd}s_{v'jd}} \delta_{vj} - \sum_{v=1}^{V_j} s_{vjd} \delta_{vj}, \quad (12)$$

where $X_{ojd} = \sum_{v=1}^{V_j} M_{ovj}x_{ovjd}$ and I assume that d is small such that a small change in income in the destination does not affect firm entry in the origin, $dM_{ovj}/dI_d = 0$. Exports increase disproportionately with destination income ($\varepsilon_{ojd} > 1$) if

$$\sum_{v=1}^{V_j} \frac{s_{ovjd}s_{vjd}}{\sum_{v'=1}^{V_j} s_{ov'jd}s_{v'jd}} \delta_{vj} - \sum_{v=1}^{V_j} s_{vjd} \delta_{vj} > 0.$$

Given the equilibrium expenditure shares in (10), I show in appendix A.2 that the condition for disproportionate exports to high-income countries can be reformulated as requiring a positive association of the relative number of firms per variety and variety quality,

$$Cov_{ojd} \left(\frac{M_{ovj}}{\bar{M}_{vj}}, \delta_{vj} \right) > 0 \quad (13)$$

if I re-interpret the variety expenditure shares, s_{vjd} , as the probability of consuming variety vj in d based on the results in Anderson et al. (1987) and Bohr et al. (2023).

¹⁰For instance, Dingel (2017) and Costinot et al. (2019) derive results from model approximations around a symmetric equilibrium.

The covariance is defined over varieties v with respect to the probability measure induced by s_{vjd} and is origin-specific since $\frac{M_{ovj}}{M_{vjd}}$ depends on the origin's firm distribution. I summarize this result in Proposition 1:

Proposition 1. *Non-unitary income elasticity. Assume that countries are small in the sense that small variations in a foreign country's income do not affect firm entry and the income elasticity of exports is given by (12) and demand given by (10). Further, interpret the variety-expenditure shares s_{vjd} as probabilities. Then, $\varepsilon_{ojd} > 1$ if the association between the scaled number of firms per variety, $\frac{M_{ovj}}{M_{vjd}}$, and its quality δ_{vj} is positive, $Cov_{jod}(\frac{M_{ovj}}{M_{vjd}}, \delta_{vj}) > 0$, else $\varepsilon_{ojd} \leq 1$.*

Proof. See appendix. □

Without variation in the share of firms across varieties, the model collapses to a version without product differentiation. In that case, there are no differential income effects on imports from different exporters. With product differentiation and a non-degenerate firm distribution across varieties, richer countries import, all else equal, more from countries with relatively more suppliers of high δ_{vj} -varieties since richer consumers prefer varieties with higher δ_{vj} . The condition is however country pair-specific due to the expenditure distribution in the destination. Differences between destinations in the expenditure distribution across varieties imply a different weighting of segments of the firm distribution in the exporting country. To illustrate, consider an exporting country with a concave firm distribution across quality levels. From the perspective of a low-income destination, which imports mostly low-quality goods, the number of firms in the exporting country is positively linked to quality. However, from the perspective of a high-income country, which spends more on high-quality varieties, the same exporter appears to have a relatively flat firm distribution. The opposite pattern holds if the firm distribution is convex.

Second, the model predicts that the firm distribution across varieties depends on a trade-off between proximity to consumers, competition and differentiation costs. The firm distribution across varieties is endogenous. The number of firms per variety is implicitly defined as a function of δ_{vj} by the zero profit condition (11). To simplify

the analysis, I assume that economies are small such that entry in country o' does not affect entry in country o . Then, by the implicit function theorem,

$$\frac{\partial M_{ovj}}{\partial \delta_{vj}} = - \frac{\partial \Delta_{ovj} / \partial \delta_{vj}}{\partial \Delta_{ovj} / \partial M_{ovj}}, \quad (14)$$

where

$$\frac{\partial \Delta_{ovj}}{\partial \delta_{vj}} = \sum_{d=1}^K (\tau_{od})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} (1 - s_{vjd}) \left[\ln(I_d) + (1 - \sigma) \frac{1}{\kappa_{vj}} \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} \right]$$

and

$$\frac{\partial \Delta_{ovj}}{\partial M_{ovj}} = \sum_{d=1}^K (\tau_{od})^{1-\xi} \eta_{jd} I_d \left(\frac{1}{\check{M}_{vjd}} \frac{s_{vjd}}{\check{M}_{vjd}} \underbrace{\frac{\partial \check{M}_{vjd}}{\partial M_{ovj}}}_{>0} \right) \underbrace{\left[\left((1 - s_{vjd}) \frac{1 - \sigma}{1 - \xi} \right) - 1 \right]}_{<0} < 0$$

with $\check{M}_{vjd} \equiv \sum_{o=1}^K (a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj}$. Locations differ in how much competition decreases market size, and in how strongly proximity to consumers valuing high-quality varieties increases market size. While the competition effect unambiguously deters entry across all varieties, a preference-cost trade-off determines the skewness in the firm distribution across varieties. The share of firms in v increases with δ only if

$$MP_{ovj} \equiv \sum_{d=1}^K \frac{(\tau_{od})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} (1 - s_{vjd})}{\sum_{d'=1}^K (\tau_{od'})^{1-\xi} \eta_{jd'} I_{d'} \frac{s_{vd'}}{\check{M}_{vd'}} (1 - s_{vd'})} \ln(I_d) > (\sigma - 1) \frac{1}{\kappa_{vj}} \frac{\partial \kappa_{vj}}{\partial \delta_{vj}}. \quad (15)$$

The left-hand side of equation (15) is a trade-cost and market size- weighted average of other countries' income and measures country o 's exposure to preferences for high-quality goods. This weighted average can be understood as an exporter's *market potential*. Higher market potential induces firm entry if the preference premium compensates for the cost disadvantage of producing higher quality goods. There exists a threshold value, $\overline{MP}_{vj}$, above which the number of firms entering a market increases with δ_{vj} . I summarize this insight in the following proposition:

Proposition 2. *Firm sorting. Assume that differentiation costs κ_{vj} are constant across origins and destinations, and firms sort across varieties such that (11) holds.*

1. $\frac{\partial M_{ovj}}{\partial \delta_{vj}}$ increases with a rise in $\frac{\partial \Delta_{ovj}}{\partial M_{ovj}}$ or a rise in $\frac{\partial \Delta_{ovj}}{\partial \delta_{vj}}$.
2. $\frac{\partial M_{ovj}}{\partial \delta_{vj}} > 0$ iff $MP_{ovj} > \overline{MP_{vj}}$, where $\overline{MP_{vj}}$ is defined by equality in (15).

Proof. Immediate from (14) and (15). $\square$

A lower sensitivity of the global market size to competition can increase $\frac{\partial M_{ovj}}{\partial \delta_{vj}}$. However, it can never change the sign of $\frac{\partial M_{ovj}}{\partial \delta_{vj}}$. Hence, the demand and competition channels can offset each other to flatten the distribution of firms across varieties but never reverse the link between the number of suppliers and the quality of varieties. Note that the firm distribution need not be monotonic. Depending on the relative convexity of the left- and right-hand side of (15), the distribution of firms across varieties can be (inverse) uni- or bimodal.

The two propositions combine to the core prediction of the model: richer countries import more from countries with higher market potential, all else equal. This prediction is what I call “neighborhood market effects”. From Proposition 1, we know that richer countries import more from countries with more firms in relatively high-end production in the region where destination d ’s expenditure is concentrated.¹¹ A factorization of the covariance shows that differences in its sign stem from the expected change of the number of firms with δ_{vj} :

$$Cov_{ojd}\left(\frac{M_{oj}(\delta_{vj})}{\widetilde{M}_{vj d}}, \delta_{vj}\right) \approx Var_{jd}(\delta_{vj}) \mathbb{E}_{ojd}\left(\left(1 - \frac{M_{oj}(\delta_{vj})}{\widetilde{M}_{vj d}} \theta_{ojd}\right) \frac{1}{\widetilde{M}_{vj d}} \frac{\partial M_{ovj}(\delta_{vj})}{\partial \delta_{vj}}\right),$$

where θ_{ojd} is defined in (8). From Proposition 2, we know that its sign is determined by the relative market potential of the exporter. Taken together, the two propositions lead to Corollary 1:

Corollary 1. *Neighborhood market effects. All else equal,*

$$\varepsilon_{ojd} > \varepsilon_{o'jd} \quad \text{if} \quad MP_{ovj} > MP_{o'vj} \quad \text{and} \quad \frac{\partial \Delta_{ovj}}{\partial M_{ovj}} = \frac{\partial \Delta_{o'vj}}{\partial M_{o'vj}}$$

and

$$\varepsilon_{ojd} > 1 \quad \text{iff} \quad MP_{ovj} > \overline{MP_{vj}}$$

¹¹The qualification is necessary because the variety expenditure share distribution is destination-specific.

around the modal variety in d .

The expenditure share distribution across varieties in the destination is critical for the sign of the covariance. Given the possibly non-monotonic relation between the number of suppliers and a variety's quality, it is important to keep in mind that bilateral trade patterns reflects the exporter's firm distribution *around the modal variety in the destination*. For instance, for the richest countries, spending is concentrated on high- δ_{vj} varieties,

$$\frac{\partial s_{vjd}}{\partial \delta_{vj}} = s_{vjd}(1 - s_{vjd}) \ln(I_d) > 0,$$

such that more weight is put on high-end varieties. A country can therefore command disproportionate import shares in high-income *and* low-income destinations if its firm distribution is convex. What is key is that any positive relation between the number of firms and the quality of the variety at any segment of the firm distribution comes only from the “proximity to demand”-mechanism, all else equal.

The next section explains the empirical strategy to test for demand effects on the location of production based on Proposition 1 and Corollary 1. The section also discusses generalizations of the theory and how they affect the empirical strategy.

3 From theory to estimation

3.1 Baseline estimation equation

Proposition 1 motivates a revealed preferences approach to shed light on exporters' comparative advantage in production of differentiated goods. The identifying assumption is that observing where a country exports to is informative about what its firms produce. Around an equilibrium without vertical product differentiation, I can express bilateral sales, up to a first-order approximation, by

$$\ln(X_{jod}) \approx \psi_{jo} + \varphi_{jd} + \gamma_j \ln(\tau_{od}) + \beta_j \ln(I_d) \text{Cov}_{jo}(M_{ovj}, \delta_{vj}) + \nu_{jod} \quad (16)$$

where each parameter has a structural interpretation as shown in the appendix B. Bilateral trade depends on a product-exporter component common to all destinations (ψ_{jo}), a product-importer component common to all exporters (φ_{jd}), and bilateral trade frictions τ_{od} . Additionally, bilateral trade is determined by the complementarity between the firm composition in o , $Cov_{jo}(M_{ovj}, \delta_{vj})$, and the demand across varieties in d . Conveniently, while Proposition 1 relies on the full vector of expenditure shares across varieties in the destination, observing real income $\ln(I_d)$ suffices for a first-order approximation. ν_{jod} is a function of the random taste draw α_{vjd} and is the structural error term. In line with Proposition 1, a positive estimate of β_j suggests that the exporting country hosts relatively more high-end producers.

Proposition 2 and Corollary 1 provide a strategy to disentangle “neighborhood market effects” as a source of comparative advantage. Under the assumption that differentiation costs are constant across exporters - a point I will discuss below - Proposition 2 states that the covariance increases with either weaker competition effects or stronger proximity to consumers valuing high-quality varieties. Yet, it becomes positive only if exposure to consumers valuing quality is above a threshold. Countries above that threshold export disproportionately to high-income destinations. “Neighborhood market effects” are thus identified from a non-linear effect of market potential on export patterns.

For an empirical measure of market potential (MP), I construct a trade cost-weighted average of destination incomes:

$$MP_{jot} = \sum_{m=1}^K \frac{\tau_{om}^{\hat{\gamma}_h} \exp(\hat{F}_{hmt})}{\sum_{m'=1}^K \tau_{om'}^{\hat{\gamma}_h} \exp(\hat{F}_{hm't})} \ln(I_{mt}), \quad (17)$$

where a “ $\hat{}$ ” denotes estimates from an auxiliary gravity regression. I estimate the weight components using bilateral trade data at the HS 6-digit level¹²:

$$\ln(X_{h,odt}) = \beta_{h,o} \ln(I_{dt}) + F_{h,ot} + F_{h,dt} + \gamma_h \ln(\tau_{od}) + \nu_{h,odt}. \quad (18)$$

¹²While the auxiliary estimation is done at the HS 6-digit level, the main regression uses HS 4-digit products. I therefore replace the weight in (17) with its across HS 6-digit goods per HS 4-digit code.

In line with equation (15), the weights combine two forces: they decrease with bilateral trade costs between origin o and market m , and they increase with destination market size $\hat{F}_{hmt}$. Trade costs $\ln(\tau_{od})$ are modeled as a linear combination of log bilateral distance, and indicators for WTO membership, common RTAs, common official language, contiguous borders, and colonial relationships. Product-market size $\exp(\hat{F}_{h,dt})$ is captured by destination-HS6 fixed effects. I_{dt} denotes real per capita GNI. For the baseline regressions, I include the full set of potential destinations $\mathcal{K}$, whether or not origin o currently exports to them, but exclude the domestic country. Instead, I control for an interaction of the exporter's domestic per capita income in the main regression. This is justified by considering that per capita income also reflects factor endowments, a supply-side determinant of product differentiation. I return to this point in the discussion in section 3.3. For robustness, I vary the set of destinations to be considered in an exporter's market potential, the weights for each potential destination, and I correct average income for inequality.

To test for third-country demand effects, I construct bins from the within-product distribution of market potential and estimate

$$\begin{aligned} \ln(X_{jodt}) = & F_{ot}^j + F_{dt}^j + \gamma^j \ln(\tau_{odt}) \\ & + \sum_{b=1}^{B=5} \beta_{b,1}^j \mathbb{1}\{o \in b\} \ln(I_{dt}) + \beta_2^j \ln(I_{ot}) \ln(I_{dt}) + \nu_{odjt}, \end{aligned} \quad (19)$$

where I introduce a time dimension (t) and a dummy variable $\mathbb{1}\{o \in b\}$ which equals one if the exporter is in quintile b of the market potential distribution. I control for real income with real GNI per capita. Finally, I parameterize trade costs τ_{odt} as a linear function of (log) bilateral distance, and dummies for common membership in a regional trade agreement, membership in the WTO, a shared language, borders, and colonial past. I estimate equation (19) product-by-product with panel OLS as the baseline method,¹³ and in line with theory, cluster standard errors by product-country

¹³Following Silva and Tenreyro (2006), much of the international trade literature proposes estimating aggregate gravity equations in multiplicative form with PPML to avoid bias from heteroskedasticity. In the case of product-level trade, the recommendations are less clear and the RESET test for model specifications does not point to PPML as the optimal model. For completeness, I discuss PPML results as a robustness check.

pair. This clustering accommodates arbitrary correlation of the error term over time and within goods due to unobserved taste for products from a particular source. I assess the robustness of my results to different ways of clustering standard errors in section 5.

The test of third-country demand effects is then whether $\beta_{1,b=b'-1}^j \leq \beta_{1,b'}^j$, with strict inequality at least for $b' = 5$, and $0 < \beta_{1,b=5}^j$, i.e. whether exports are biased towards high-income countries when market potential is high enough. The idea is closely related to the test for a ‘weak home market effect’ in Costinot et al. (2019). The key difference is that while the ‘weak home market effect’-test tests for higher *levels* of bilateral exports from countries with a positive domestic demand shock for a given product, the test for third-country effects tests for systematic *biases* in exports across destinations, *ceteris paribus*.

3.2 Identification

The rich set of fixed effects and trade cost controls isolate the role of the demand-supply complementarity from further trade determinants. The structural interpretation of the fixed effects and parameters help clarify identification. Exporter-product-year effects, F_{ojt} , capture the country’s average supply capacity for a given product through domestic factors like productivity and wage levels that determine a country’s Ricardian comparative advantage at the product level. Beyond the model structure, these fixed effects also control for the *average* quality level of exports across destinations, which has been shown to increase with market potential in Lugovskyy and Skiba (2015) and Dingel (2017).

The destination-product-year effects, F_{djt} , in turn, control for a product’s market share in the destination and the country’s supplier access. Thus, the fixed effects account for a positive correlation between destination income and access to highly productive suppliers in general. If preferences were homothetic, the destination effects would absorb destination per capita income and β_j could not be identified. Beyond the structural interpretation, the fixed effects also capture the average level of quality imported in the country, which shifts the bilateral income elasticity as seen in (12).

With both fixed effects in place, variation in bilateral trade flows identifies the coefficient of interest β_j . The trade cost function τ_{odt} removes variation from geographic, cultural and monetary bilateral trade barriers/facilitators. Accounting for bilateral trade costs is critical to not confound third-country demand effects with ‘direct’ demand effects due to proximity of the two trade partners. An alternative to relying on trade cost proxies is to introduce country-pair fixed effects in line with the recommendation by Yotov et al. (2016) and Larch et al. (2025). While this is feasible, the coefficients of interest are then only recovered from yearly variation in market potential and destination per capita income. Since the theoretical intuition derives from cross-sectional variation, saturating the model with country-pair fixed effects seems restrictive, and I will only use such a specification as robustness check.

For consistency, destination per capita income and exporter market potential must not be correlated with the error term given the set of controls. As such, the covariance between tastes shocks across varieties in the destination and the firm distribution in the origin must be uncorrelated with both the destination’s income and the exporter’s market potential.

Together, the control variables absorb a wide array of endogeneity sources. Even so, reverse causality from trade to income per capita remains a concern. Note that the aggregate effect of trade on per capita income (independent of the trade partner) is partialled out by the fixed effects. A reverse effect could bias the coefficients $\{\beta_1\}_b$ and β_2 if trade in a given product affects importer per capita income differentially depending on the origin of the product. For instance, if the theory of section 2 is applied for intermediate goods, and countries with higher market potential sell goods of higher quality, the higher quality inputs could make the importer more productive and raise its per capita income. Reverse causality in this way would upward bias the estimate. Reverse causality is less likely however to affect trade in consumption goods. Relatedly, trade could have a reverse effect on market potential if trade influences whom to integrate with, e.g. through regional trade agreement. While this effect is unlikely to occur at the product level, I address the concern by exploring the dynamics of the market potential effect.

3.3 Theoretical extensions and empirical robustness

The model in Section 2 intentionally keeps firm behavior simplistic. The idea for the theoretical analysis is to focus fully on the demand-side mechanism induced by market access. Empirically, however, several supply-side factors could confound or attenuate the relation between bilateral trade and third-country demand.

Designing-to-market. Consider the possibility that firms have the option to sell multiple varieties, and ‘design to market’ such that $\delta_{vjd} \neq \delta_{vjd'}$. A similar setting is discussed in Lugovskyy and Skiba (2015) (multi-quality producers) and in numerous studies with firm-level data (e.g. Bastos and Silva (2010); Brambilla et al. (2012); Manova and Zhang (2012); Flach (2016); Manova and Yu (2017)). In appendix A, I introduce designing-to-market through heterogeneity in marginal costs $\kappa_{vjd} \neq \kappa_{vjd'}$. For instance, firms may use different inputs for different destinations. In this case, third-country demand still shifts bilateral trade though the income-cost trade-off (15) that determines the firm selection across quality levels has to be solved per destination. More firms offer high-quality varieties if the trade-off is favorable in sufficiently many destinations. Alternatively, if market segmentation comes at a destination-specific fixed cost f_{jd} , a zero profit condition will need to hold for each market separately. In this case, the firm distribution across quality levels is determined by selection into serving a given destination market. However, selection for one country does not affect firm selection for another country and hence bilateral trade should be independent of third-market demand.

Pricing-to-market. Related to the ‘designing-to-market’ question, consider the case of ‘pricing-to-market’. Knowing that richer consumers have a higher willingness to pay for high-quality goods, firms in the high-quality segment could charge higher prices in richer countries without incurring additional costs. Empirically, Simonovska (2015), among others, provides evidence for such pricing patterns. Pricing-to-market will downward bias the coefficient of interest β_j since richer consumers buy less than what they would have if prices were constant. Empirically, I address both concerns, designing-to-market and pricing-to-market, by constraining the sample such that the

variation of unit values within exporter-products across destinations is small.

Heterogeneous factor endowments. In addition, consider cross-country differences in high-skilled labor endowment. Caron et al. (2014) show that countries abundant with high-skilled labor have a comparative advantage in supplying high-quality goods. In my model, I could introduce such differences by allowing for heterogeneity in differentiation costs κ_{vj} across locations o . Such heterogeneity would change producer prices and benefit importers in proximity to high-skill-endowed countries by lowering their price indices. Empirically, the former effect is absorbed by the exporter-product fixed effect. Critical for identification is that cross-country variation in the number of firms per quality level can no longer be uniquely attributed to third-country demand effects. The covariance between the number of firms and the quality level of a variety can be positive if differentiation costs are low enough (see equation (15)). To address the concern, I control for domestic per capita income interacted with destination per capita income separately and construct market potential to be based on foreign income only.

Global value chains. Next, consider the role of global value chains and in particular the outsourcing of final production stages by firms in high-income countries. The model is silent about the origin of the mass of firms present in a given location. It only describes the distribution of firms across quality segments. As such, if outsourcing destinations are chosen based on the access to high-income consumers, the model mechanism holds and conclusions about the role of foreign demand for high quality production remain valid. If, however, the outsourcing destination is chosen based on supply-side factors such as factor endowments and productivities, and if those are correlated with the country's proximity to high-income markets, the coefficient in the baseline model would be biased. Adding an interaction between exporter per capita income and the importer's per capita income also absorbs this channel.

Wage effects. Finally, market potential might indirectly affect an economy's output and trade patterns through its effect on wages as discussed in Breinlich and Cuñat

(2013).¹⁴ Omitting the exporter’s per capita income would induce a bias away from zero on the coefficient β_1 through the wage channel.

4 Data

4.1 Trade data

This study makes extensive use of data provided by Centre d’Etudes Prospective et d’Informations Internationales (CEPII). I obtain data on yearly, bilateral trade values at the 6-digit product level of the Harmonized System (version 1996) for the years 1999-2019 (*BACI* database, Gaulier and Zignago (2010)). Because the theory applies for trade in differentiated goods, I extract data on consumption and intermediate goods only. I aggregate the yearly data to 3-year periods, and the 6-digit codes to 4-digit products to limit measurement error and reduce the number of zero trade flows. Further, I impose constraints on the country and goods included as follows. First, countries must have a population of more than 1 million in any given period. Second, the world trade value of a good needs to be at least 1 billion US dollars. Finally, each exporter-product-period and each importer-product-period must appear a minimum of 15 times. The main sample consists of 140 destinations, 139 exporter countries, and 436 HS4-goods. 275 of these are consumption goods, 111 classify as intermediate goods, and 50 HS4-codes encompass HS6-codes that are classified as both consumer and intermediate goods. In total, the sample has more than 11 million observations. The sample includes substantially more countries than most other studies on this topic, e.g. Hallak (2006) covers only 60 countries while Lugovskyy and Skiba (2015) observe imports of only seven Latin American countries.

Regarding control variables, I capture trade costs with data on bilateral distance and indicators about a shared border, colonial past and common language from the *Gravity* database (Version 02/2022) also made available by CEPII. This data set also contains information on WTO membership of the trade partners as well as GDP per

¹⁴A higher market potential implies elevated demand for labor, which can put upward pressure on wages. In the model, this effect is ruled out by the presence of the homogeneous outside good, which accommodates any changes in labor demand and keeps wages fixed.

capita (PPP) and population values originally reported by the World Development Indicator database of the World Bank. Information on regional trade agreement participation comes from *Mario Larch's Regional Trade Agreements Database* (Egger and Larch, 2008). For a robustness exercise, I obtain Gini data from the *World Income Inequality Database*. This database is a collection of Gini data from primary sources. I choose the Gini index defined in terms of income inequality (as opposed to consumption inequality) and keep values referring to the total population and covering both urban and rural areas in a country. I then take averages of duplicate entries.

Table 1 provides descriptives for key trade statistics, separately listing averages for high-income countries and all other countries. The sample split highlights the importance of global inequality for trade: high-income countries on average import almost three times as much as other countries in values and import close to twice as many different products.

Table 1: Summary statistics of trade margins

	Full sample		HIC Importer	Other Importer
	Mean	Sd	Mean	Mean
trade value	6,939	199,445.52	13,131	2,642
Number imported HS 4	269	127.26	388	233
Obs. in Mio.	11.5			

Notes: Trade values in 1000s.

4.2 Empirical market potential

Table 2 describes the distributions of foreign market potential, and per capita income of both importer and exporter in the sample. Relative to the GDP per capita distribution, the global distribution of market potential is more compressed and shifted upwards, as evident in Table 2. Yet, the magnitudes of market potential and GDP per capita are comparable and variation in market potential is high. Table 2 also reveals the wide development spectrum covered in the sample. The poorest country in the sample is Mozambique and the richest is Singapore.

Table 3 presents the raw correlation of market potential and per capita income with further country characteristics. Richer and larger countries tend to have a higher

Table 2: Summary statistics for main independent variables

	Mean	Sd	25th percentile	Median	75th percentile	Min	Max	N
Importer GDP per capita	8.05	1.66	6.74	8.07	9.33	0.76	11.50	1,178
Exporter GDP per capita	8.15	1.63	6.86	8.14	9.41	0.76	11.50	1,120
FMP _{jot} , bin 1	7.34	2.28	6.51	7.99	8.98	0.78	10.73	38,152
FMP _{jot} , bin 2	8.50	1.55	8.05	8.90	9.52	1.06	10.78	37,893
FMP _{jot} , bin 3	9.21	0.83	8.77	9.36	9.80	2.05	10.81	37,875
FMP _{jot} , bin 4	9.63	0.63	9.27	9.73	10.07	2.34	10.86	37,896
FMP _{jot} , bin 5	10.09	0.53	9.84	10.20	10.46	3.10	11.23	38,069

Notes: Data in logs. Bins mark qunitiles of the market potential distribution within 4-digit HS codes.

market potential. The correlation between market potential and domestic per capita income is however low enough to not be concerned of multi-collinearity in the regression. Countries with higher market potential tend to sell to richer destinations, a first hint that the income elasticity of trade is indeed a function of market potential. Distance to trade partners is weakly negatively linked to market potential. Finally, average unit values - whether raw or purged of variation in trade cost proxies and importer-product-year effects - are positively correlated with both income per capita and market potential.¹⁵ The positive though small correlation of market potential and unit values echoes the findings on the role of geography of income for average unit values in Lugovskyy and Skiba (2015) and Dingel (2017) discussed above. I analyze unit values in more detail in section 5.3.

Table 3: Correlation table

	log GDPpc _{ot}	log GDP _{ot}	log GDPpc _{dt}	log av. dist _{ot}	wght. av. log unit values _{jot}	wght. av. residualized unit values _{jot}
FMP _{jot}	0.242	0.065	0.092	-0.043	0.200	0.074
GDPpc _{ot}	1.000	0.258	-0.010	-0.028	0.185	0.303

Notes: Raw correlation coefficients. Unit values: weighted average hs4 unit values with weights the destination-hs4 share in exporter-hs4-year trade. Unit values residualized from distance, contiguity, common language, common colonial past and importer-hs4-year effects.

5 Results

5.1 Main results

Table 4 and Figures 1 and 2 summarize the main results. Table 4 reports the summary statistics of the coefficient estimates of equation (19) for the 436 HS 4-digit product

¹⁵The unit values are computed at the HS 6-digit level and then aggregated to the HS4-level with weights the trade share of a HS6-importer in HS4-exporter-year sales.

codes. The top panel summarizes the coefficients related to the interaction terms with destination per capita income while the bottom panel summarizes the coefficients related to the trade cost proxies. The last column lists the number of products for which the coefficient estimates are statistically different from 0 at the 95% significance level.

The estimates are largely consistent with the model predictions. In particular, evidence for a non-linear effect of market potential on exports to high-income countries is strong. This is both true in terms of sign and magnitude of the coefficients and number of products with statistically significant effects. A Wald test of joint nullity cannot be rejected for 70 products at the 5% significance level. The model suggests that for these products the cost and competition channels outweigh the demand side incentive to produce high quality goods. The additional independent variables affect bilateral exports as expected.

The non-linear effect of market potential predicted by the model is clearly visible in Figure 1. Panel (a) plots the coefficient estimates and corresponding confidence bounds from estimating equation (19) on a pooled sample of products. The coefficient becomes positive and significant only for countries in the top quintile of the market potential distribution. Panel (b) shows averages of the coefficient estimates and contrasts results for the full model with a “naive” specification that does not control for interactions of per capita income with exporter per capita income. Though the magnitude of the coefficients decline on average in the full model specification, the non-linearity of coefficient estimates is strong and present in both cases.

Figure 2 shows the distribution of coefficient estimates by HS chapter for countries in the second quintile and countries in the top quintile of the market potential distribution. Countries in the bottom quintile are the reference group. The figure again reinforces the non-linearity result. For majority of the 14 different HS chapters, the median coefficient in the second market potential quintile is at or below zero. In contrast, the median coefficient for the countries in the top quintile is above zero for all HS chapters except leather and foodstuffs. In terms of the model, a negative estimate on the top quintile market potential suggests that competition and cost factors deter

entry in high-end production despite proximity to demand for high-quality goods.

Table 4: Summary statistics for coefficient estimates, main regression

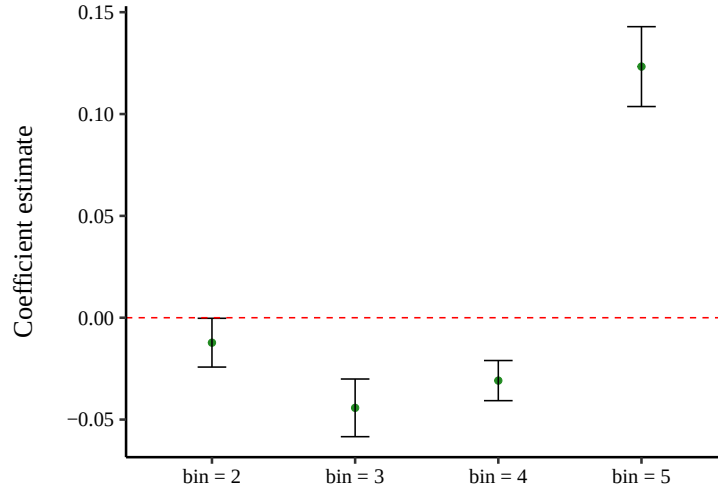
	Mean	SD	p25	p50	p75	# HS4 sig.
FMP bin 2	0.02	0.18	-0.12	0.07	0.15	194
FMP bin 3	0.01	0.23	-0.18	-0.08	0.18	220
FMP bin 4	0.05	0.28	-0.17	0.11	0.25	215
FMP bin 5	0.22	0.29	0.13	0.26	0.36	250
$\log(y_o)$	0.03	0.10	-0.05	0.06	0.10	304
$\log(\text{distance})$	-1.32	0.30	-1.53	-1.31	-1.13	436
colonial ties	0.66	0.20	0.52	0.64	0.78	361
common language	0.75	0.25	0.57	0.73	0.91	401
contiguity	0.88	0.28	0.71	0.86	1.03	424
RTA	0.33	0.17	0.22	0.29	0.40	297
WTO	1.13	0.53	0.82	1.03	1.28	255
Number of HS4 codes						436

Notes: Summary statistics for parameter estimates. The HS4-level regressions control for exporter-year and importer-year fixed effects. The last column shows the number of products for which the parameter estimates are significantly different from 0 at the 5% confidence level. We cannot reject the null hypothesis of joint nullity of market potential bins for 70 products. The total number of products is 436.

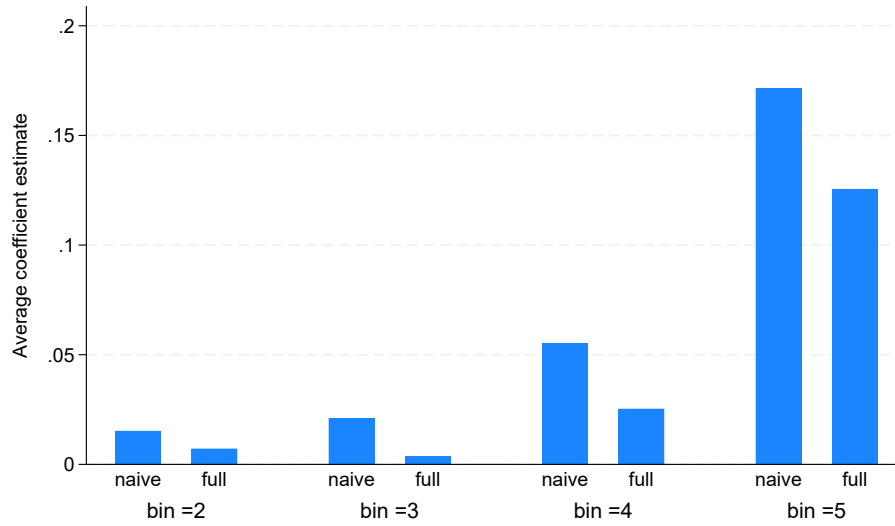
The baseline results are robust to alterations in the estimation strategy as appendix D details. First, I change the estimation method to employ PPML estimation (Silva and Tenreyro, 2006). The coefficients are strongly positively correlated and the non-linear pattern persists. However, which products show significant effects changes somewhat between PPML and OLS estimation. Second, I control for country-pair fixed effects instead of working with trade cost proxies. This is a rather demanding specification and results in much fewer significant coefficients than in the baseline. Nonetheless, the non-linearity in the market potential effect shows also in these estimation results. Third, I change the clustering of standard errors and find that the share of products with significant coefficients roughly halves when standard errors are clustered by exporter and importer instead of by country-pair. This suggests allowing for error correlation across exporter-years (importer-years) by importer (exporter) makes estimates more noisy. Fourth, I change the empirical measure of market potential, both in terms of construction and time of measurement. In terms of constructing market potential, I explore changes to the set of countries considered for the “global” market access by restricting the set to neighboring countries or trade partners only. Moreover, I partial out within-country inequality effects from per capita income be-

Figure 1: Main results - Average effects across products

(a) pooled product sample

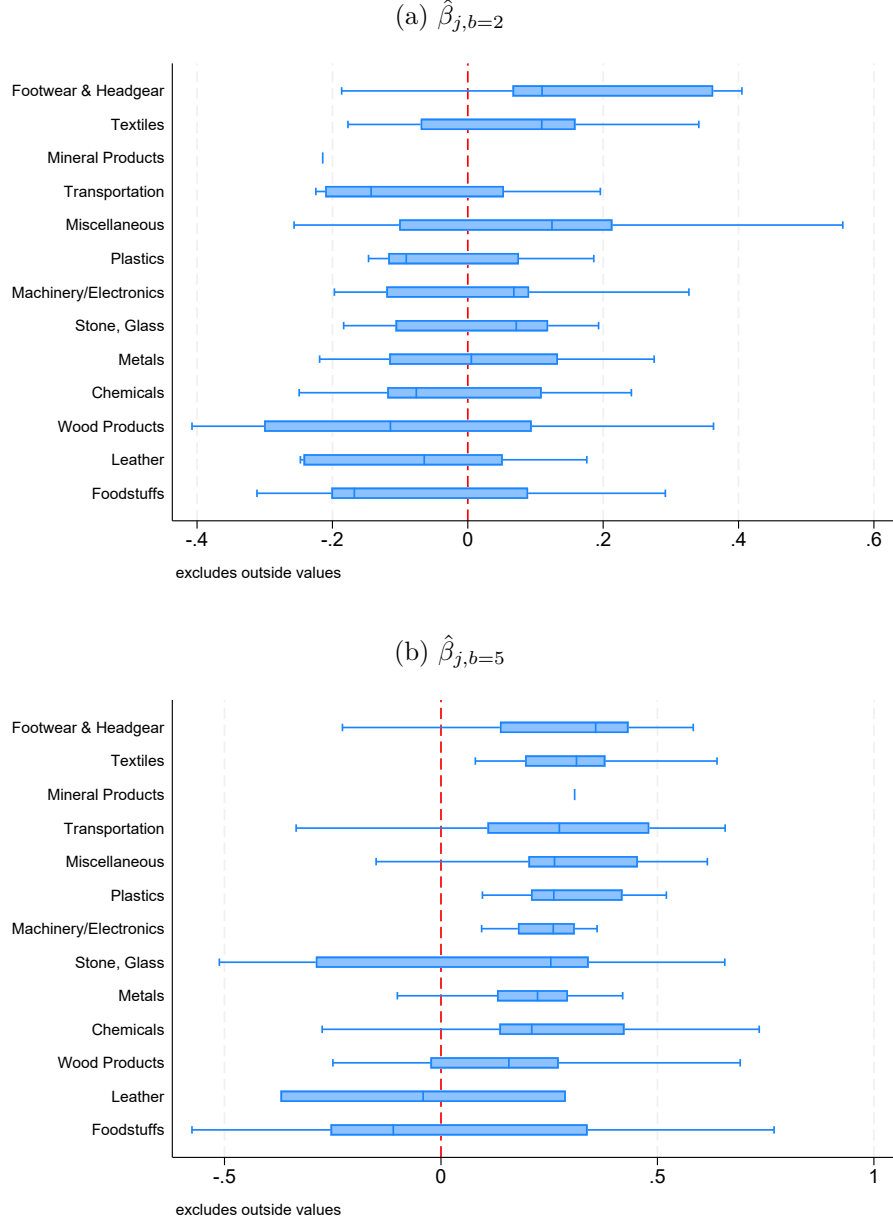


(b) averages across products



Notes: Panel (a) shows the coefficient estimates of $\beta_{j,b}$ with 95%-confidence bounds from estimation of (19) on a pooled product sample. Panel (b) shows the average estimates across products for a “naive” version of (19), which does not control for exporter per capita income, and the full model. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for calculations in panel (b).

Figure 2: Main results - Distribution of effects across products



Notes: Panel (a) shows box plots of the coefficient estimates of $\beta_{j,b=1}$ from estimation of (19). Panel (b) shows box plots of the coefficient estimates of $\beta_{j,b=5}$. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for calculations. Outliers are dropped from the plots. The solid line in the middle of each box plot marks the median per HS chapter.

fore constructing market potential. Finally, I change the factors entering the weights assigned to each market by either constructing them from observed data instead of trade cost and market size estimates, or by employing data including domestic trade in the estimation of market size. Either alternative weight is more aggregate than the baseline version and results deviate somewhat from the baseline. Similarly, estimates based on a market potential considering only neighbors as the “global market” also deviate from the baseline. Fifth, I compute market potential bins from countries’ lagged market potential. The results of this version are highly correlated with the baseline results.

5.2 Model consistency checks

Having established the robustness of the baseline results to alternative empirical approaches, I now examine more closely the consistency of the empirical results with the model assumptions.

First, I split the estimates by product group, i.e. final vs. intermediate goods using the UNCTAD definition. Reverse causality - an increase in destination income due to imports from high market potential countries - is more likely at play for intermediate goods for which we could then expect higher coefficients on the destination income interaction terms. Table 5 shows summary statistics of the results, decomposed by product group. Comparing final and intermediate goods, I find that the non-linearity is less pronounced for intermediate goods, whether assessed in terms of the coefficient magnitudes or their significance. Importantly, coefficient estimates for intermediate goods are on average lower than for final goods, which speaks against reverse causality.

Table 5: Summary statistics of baseline estimates by product groups

	FMP bin 2		FMP bin 5		$\hat{\beta}_{1,5} - \hat{\beta}_{1,2}$	share HS 4, reject $\hat{\beta}_{1,1} = \dots = \hat{\beta}_{1,5} = 0$	# hs4 codes
	mean	share sig. HS4	mean	share sig. HS4			
final goods	0.01	0.47	0.15	0.61	0.15	0.87	275
intermediates	-0.01	0.35	0.05	0.44	0.08	0.74	111
mix	0.03	0.50	0.17	0.68	0.15	0.90	50

Notes: Estimates from baseline regression model. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for statistics calculation of means.

Second, I turn to the key modelling assumption that firms pool profits across destinations to cover fixed costs of production. This pooling of profits forges a link between

potential demand in different markets and gives rise to the role of market potential. While the profit pooling assumption goes against previous findings in the literature, I present suggestive evidence for scope of market potential to matter empirically.

To assess how wide-spread destination market pooling is in the data, I compute unit values at the HS 6-digit level. I regress the log unit values on exporter-HS 6- year effects and partial out bilateral trade cost proxies (log distance, dummies for colonial past, contiguity, common language and RTA membership):

$$\log uv_{odt,hs6} = \gamma^{uv} \tau_{odt} + f_{ot,hs6} + \varepsilon_{odt,hs6}^{uv} \quad (20)$$

The variance of the residual $\varepsilon_{odt,hs6}^{uv}$ across destination-years by exporter-product indicates to what extent prices vary across destinations beyond trade costs.

To combine the analysis on unit values with the main regression, I compute the weighted-average unit value residual at the HS 4-level. I then calculate the standard deviation of the residualized unit values at the exporter-HS4 level across destination-years:

$$sd_{oj} \left(\sum_h (X_{hodt}/X_{jodt}) \hat{\varepsilon}_{odt,hs6}^{uv} \right) \equiv sd_{oj} (\tilde{u}_{jodt}).$$

Figure 3 shows the distribution of $sd_{oj} (\tilde{u}_{jodt})$ across exporters for the nine products roughly at the deciles of the distribution of the weighted-average standard deviation by product:

$$\bar{sd}_j = \sum_o (X_{jo}/X_j) sd_{oj} (\tilde{u}_{jodt}).$$

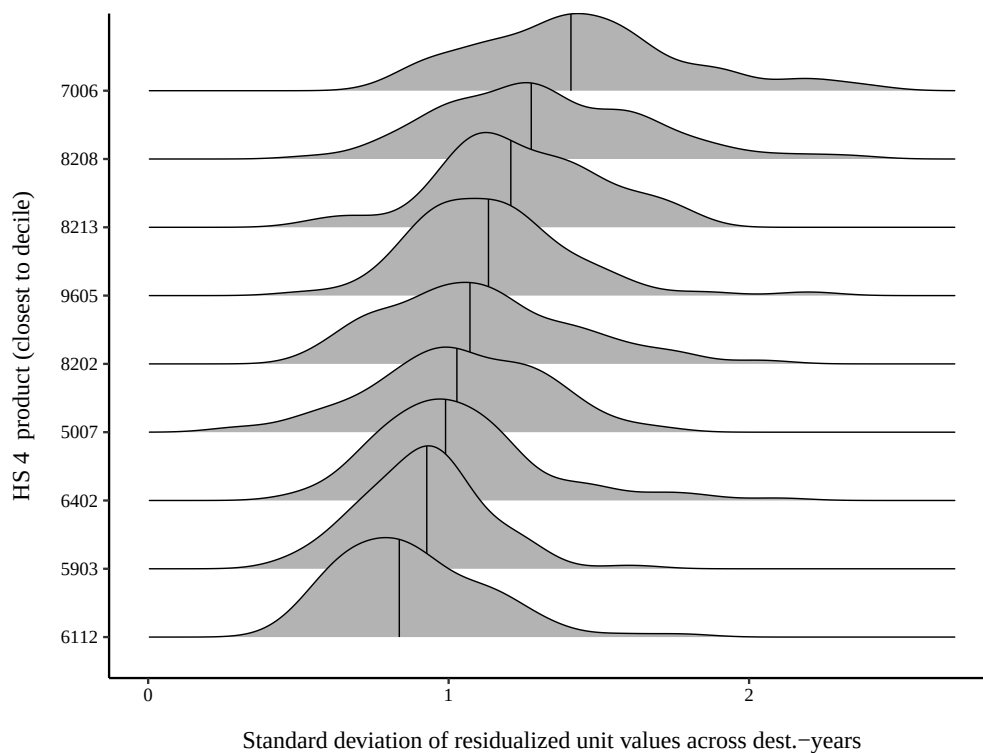
Figure 4 shows the distribution of $sd_{oj} (\tilde{u}_{jodt})$ across products for the nine countries roughly at the deciles of the distribution of the weighted-average standard deviation by country:

$$\bar{sd}_o = \sum_j (X_{jo}/X_o) sd_{oj} (\tilde{u}_{jodt}).$$

Table 8 in the appendix provides the corresponding summary statistics. The plots reveal substantial heterogeneity in how much residualized unit values vary across destinations. The distinction is more notable for different products, but also countries differ in how much their export unit values vary across destinations. Based on the

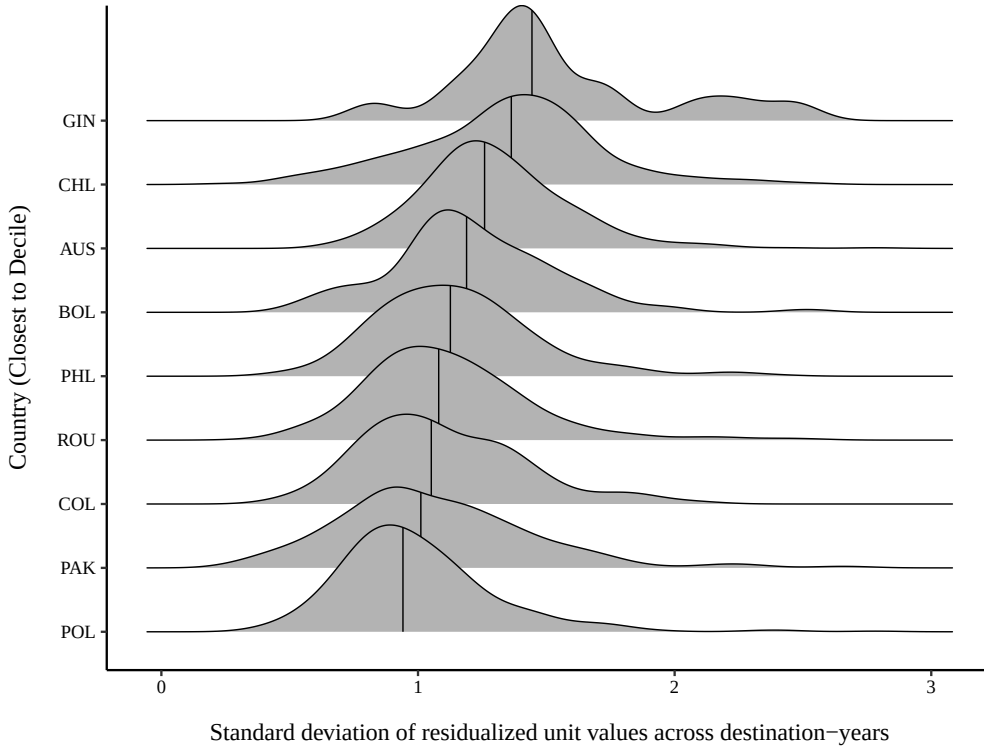
model, my expectation is that market potential has a stronger effect for exporter-products that are situated in the bottom half of the distribution per product, or per exporter, respectively.

Figure 3: Unit value variance distribution, by product across exporters



Notes: Solid vertical black line marks the median of the product-specific distribution of the standard deviation of unit values across exporters.

Figure 4: Unit value variance distribution, by exporter across products



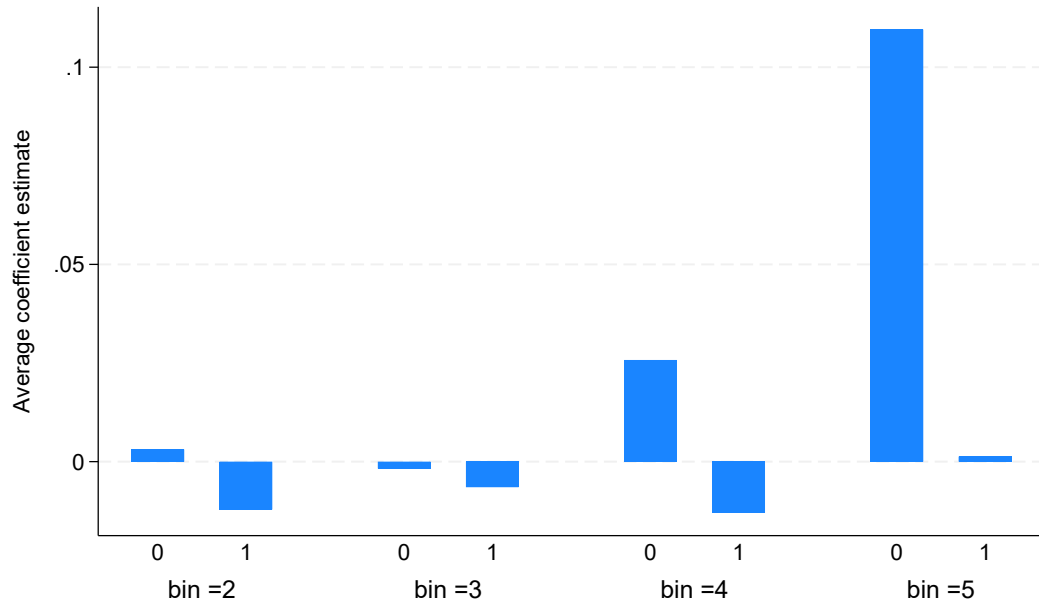
To test for the role of market segmentation, I construct two dummy variables. The first indicator equals 1 if an exporter's cross-destination variance of unit values for a given product is higher than the average variance for that product. The second indicator equals one if the exporter's cross-destination variance of unit values for a given product is higher than the average variance for this exporter. I interpret values of one as indicators for market segmentation. In line with the model, I find that market potential has stronger effects for exporters that engage in market pooling (understood as a low unit value variance). The two panels in Figure 5 shows average estimates of the tripple interaction terms. When unit values vary strongly across destinations, exporters across all market potential groups export less as destination income rises.

An alternative explanation is that pricing-to-market biases the market potential coefficient downward. If exporters systematically charge higher prices in richer countries, imports appear lower than they would be under uniform pricing. If firms from high-market-potential countries engage in pricing-to-market, demand from rich countries is

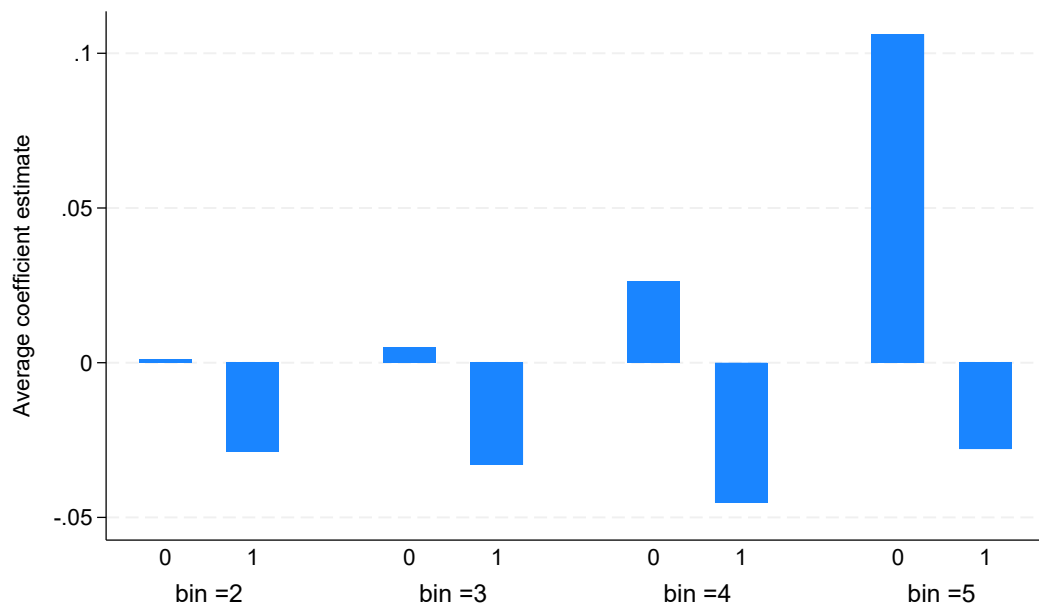
depressed, reducing the estimated effect.¹⁶

Figure 5: Heterogeneity results by degree of market segmentation

(a) $1(sd_{oj} \geq \bar{sd}_o)$



(b) $1(sd_{oj} \geq \bar{sd}_j)$



Notes: Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for calculations. The axis labels "0" and "1" refer to the condition stated in the panel title.

¹⁶If all exporters set higher prices in rich countries, this effect is absorbed by destination-product fixed effects.

5.3 A price view of the market access effect

While the present paper studies the effect of demand on the location of production from trade values - in line with the home market effect literature (Costinot et al., 2019), a number of studies on product quality infer a location's comparative advantage from unit values (Khandelwal, 2010; Lugovskyy and Skiba, 2015; Dingel, 2017; Kong, 2025). To connect to these studies, I derive the link between market access and export prices from my model. Up to a first-order approximation, my model predicts that bilateral shipment prices can be expressed as

$$\log(P_{jodt}) = \phi_{jdt} + \beta_j Cov_{jot}(M_{ovjt}, \delta_{vj}) + \nu_{jodt} \quad (21)$$

where ν_{jodt} is a function of exporter productivity and wages, the variance of firms across varieties, and bilateral trade costs. The prediction is similar as for the gravity model: prices increase with market access only if market access is strong enough to outweigh the counteracting cost and competition determinants of firm sorting. The difference is that the model does not predict a bias in prices across destinations since pricing to market has been ruled out.

The idea behind the “price view” of the home market effect is that, conditional on market shares, high-quality suppliers can charge higher prices. Typically, studies analysing locational comparative advantage have access to firm-level data such that unit values are a fairly accurate proxy for prices and factor usage is observed. With only aggregate data at hand, I can control only for exporter characteristics at the country-level. In addition, since market access is predicted to change the level of prices uniformly across destinations, I cannot control for exporter fixed effects.

Bearing these caveats in mind, I approximate prices with unit values at the HS 6-digit level and estimate the following model:

$$\log(uv_{podt}) = F_{jdt}^{uv} + \gamma^j \ln(\tau_{od}) + \sum_b \beta_{b,1}^j \mathbb{1}\{o \in b\} + \beta_2^j \ln(I_{ot}) + \nu_{odjt}^{uv} \quad (22)$$

where p denotes an HS 6-digit code and j continues to denote HS 4-digit codes. F_{jdt}^{uv} denote destination-HS 4-year fixed effects and trade costs are parameterized as in the

main regression. ν_{odjt}^{uv} is the unobserved error term.

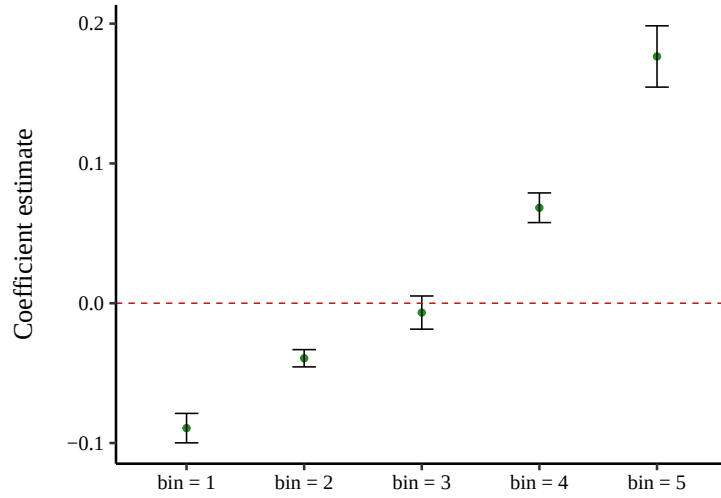
The results of a pooled regression of (22) and product-wise regressions are reported in Figure 6 and Table 6. In line with theory, the average effect of market access on prices is non-linear and increasing. However, being a country in the highest market potential quintile has a significant effect for much fewer products than the lower bins have.

Table 6: Summary statistics for coefficient estimates, unit value regression

	Mean	SD	p25	p50	p75	# HS4 sig.
FMP bin 1	-0.22	0.32	-0.39	-0.24	-0.11	348
FMP bin 2	-0.18	0.26	-0.34	-0.18	-0.10	353
FMP bin 3	-0.14	0.20	-0.25	-0.15	-0.06	341
FMP bin 4	-0.04	0.19	-0.17	-0.08	0.09	320
FMP bin 5	0.18	0.18	0.06	0.14	0.26	95
$\log(y_o)$	0.21	0.12	0.12	0.19	0.26	433
$\log(distance)$	0.13	0.10	0.07	0.12	0.17	394
colonial ties	-0.07	0.19	-0.15	-0.09	0.07	139
common language	-0.10	0.11	-0.15	-0.10	-0.07	239
contiguity	-0.14	0.15	-0.19	-0.13	-0.09	263
RTA	0.03	0.12	-0.05	0.06	0.10	232
WTO	-0.13	0.43	-0.27	-0.09	0.13	265
Number of HS4 codes						407

Notes: Summary statistics for parameter estimates of unit value regression. Statistics are computed based on significant estimates at the 95% confidence level. The last column shows the number of products for which the parameter estimates are significantly different from 0 at the 95% confidence level. We cannot reject the null hypothesis of joint nullity of market potential bins for 11 products. The total number of products is 407.

Figure 6: Unit value results



Notes: Plot shows the coefficient estimates of $\beta_{b,1}$ with 95%-confidence bounds from estimation of (22) on a pooled product sample.

6 Conclusion

This paper examines proximity to high-income countries as a driver of product differentiation. The analyses of product-level bilateral trade flows show that exports rise faster with destination income for countries with high market potential. The empirical results are consistent with a model in which firms exploit economies of scale by balancing consumer demands from many countries in their choice of product differentiation.

The results point to the potential of economic integration between countries of different per capita income to affect the countries' production through a change in the composition of consumer preferences. The analysis thus also points to an additional channel through which protectionist measures from high-income countries can have adverse consequences, notably for developing countries. However, this is conditional on production costs and competition. Particularly remote locations relative to the center of demand for a good might need to invest a lot in lowering production costs to induce production. At the same time, only lowering production costs without access

to (sufficient) demand might not suffice to induce production.

Use of Generative AI

During the preparation of this work the author used Claude Sonnet and refine.ink in order to ensure code for the empirical analyses is self-contained and logical fallacies in the text removed. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

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A Generalized model

This section presents a generalized version of the baseline model including derivations of key equations.

A.1 Model setup and equilibrium conditions

As in the baseline, consumers minimize expenditure subject to utility given by

$$\begin{aligned}\bar{U}_d = C_d &= C_{0d}^{\eta_{0d}} \prod_{j=1}^J C_{jd}^{\eta_{jd}}, \\ C_{jd} &= \left[\sum_{v=1}^{V_j} \tilde{\lambda}_{vjd}^{1-\rho} C_{vjd}^\rho \right]^{\frac{1}{\rho}}, \\ C_{vjd} &= \left(\sum_{o=1}^K \int_{\omega_o \in \Omega_{ovj}} c_{vjd}(\omega_o)^\zeta d\omega_o \right)^{\frac{1}{\zeta}}\end{aligned}\tag{23}$$

where

$$\tilde{\lambda}_{vjd} = \frac{\alpha_{vjd} \lambda_{vjd}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} \lambda_{v'jd}} \quad \text{with} \quad \lambda_{vjd} = \bar{U}_d^{\delta_{vj}},$$

and substitution parameters $\rho = \frac{\sigma-1}{\sigma}$ and $\zeta = \frac{\xi-1}{\xi}$ with $\xi > \sigma > 1$. The first order condition of the expenditure minimization problem is given by

$$\mu_d \eta_{jd} \frac{\bar{U}_d}{C_{jd}} C_{jd}^{1-\rho} \tilde{\lambda}_{vjd}^{1-\rho} C_{vjd}^{\rho-\zeta} c_{vjd}(\omega_o)^{\zeta-1} = p_{vjd}(\omega_o)$$

First, derive the price indices: Take to the power $\frac{\zeta}{\zeta-1}$, integrate over $\omega \in \Omega$ and take to the power $\frac{\zeta-1}{\zeta}$ to get the price index P_{vjd} :

$$\mu_d \eta_{jd} \frac{\bar{U}_d}{C_{jd}} C_{jd}^{1-\rho} \tilde{\lambda}_{vjd}^{1-\rho} C_{vjd}^{\rho-\zeta+\zeta-1} = \left(\int_{\omega_o \in \Omega_{ovj}} p_{vjd}(\omega_o)^{\frac{\zeta}{\zeta-1}} d\omega \right)^{\frac{\zeta-1}{\zeta}} \equiv P_{vjd}.$$

Take to the power $\frac{\rho}{\rho-1}$, multiply with $\tilde{\lambda}_{vjd}$, sum over varieties and take to the power $\frac{\rho-1}{\rho}$,

$$\mu_d \eta_{jd} \bar{U}_d C_{jd}^{-1+1-\rho+\rho-1} = \left(\sum_{v=1}^{V_j} \tilde{\lambda}_{vjd} P_{vjd}^{\frac{\rho}{\rho-1}} \right)^{\frac{\rho-1}{\rho}} \equiv P_{jd}.$$

Multiply with C_{jd} and sum over goods j ,

$$\bar{U}_d = \left(\sum_{j=1}^J C_{jd} P_{jd} \right) / \mu_d = E_d / \mu_d. \quad (24)$$

Finally, since $\bar{U}_d = C_d$, and $E_d = P_d C_d$, it follows that $\mu_d = P_d$. Absent savings, utility is proportional to real income (which equals expenditure) and I replace utility with real income from now on.

From the price equations, we can solve for the three consumption indices:

$$C_{jd}^{-\rho} = (\eta_{jd} I_d)^{-\rho} P_{jd}^{\rho}$$

$$C_{vjd} = \left[(\eta_{jd} I_d)^{1-\rho} P_{jd}^{\rho} \tilde{\lambda}_{vjd}^{1-\rho} P_{vjd}^{-1} \right]^{\frac{1}{1-\rho}}$$

$$c_{vjd}(\omega_o) = p_{vjd}(\omega_o)^{\frac{-1}{1-\zeta}} C_{vjd}^{\frac{\rho-\zeta}{1-\zeta}} \tilde{\lambda}_{vjd}^{\frac{1-\rho}{1-\zeta}} C_{jd}^{\frac{-\rho}{1-\zeta}} (\eta_{jd} I_d)^{\frac{1}{1-\zeta}}$$

$$\begin{aligned} c_{voj}(\omega) &= p_{vjd}(\omega_o)^{\frac{-1}{1-\zeta}} \left(\left[(\eta_{jd} I_d)^{1-\rho} P_{jd}^{\rho} \tilde{\lambda}_{vjd}^{1-\rho} P_{vjd}^{-1} \right]^{\frac{1}{1-\rho}} \right)^{\frac{\rho-\zeta}{1-\zeta}} \tilde{\lambda}_{vjd}^{\frac{1-\rho}{1-\zeta}} \left((\eta_{jd} I_d)^{-\rho} P_{jd}^{\rho} \right)^{\frac{1}{1-\zeta}} (\eta_{jd} I_d)^{\frac{1}{1-\zeta}} \\ &= p_{vjd}(\omega_o)^{\frac{-1}{1-\zeta}} \left(P_{vjd}^{\frac{1}{\rho-1}} \right)^{\frac{\rho-\zeta}{1-\zeta}} \tilde{\lambda}_{vjd}^{\frac{1-\rho}{1-\zeta} + \frac{\rho-\zeta}{1-\zeta}} (P_{jd})^{\frac{\rho}{1-\zeta} + \frac{\rho}{1-\rho} \frac{\rho-\zeta}{1-\zeta}} \\ &\quad (\eta_{jd} I_d)^{\frac{1}{1-\zeta} + \frac{\rho-\zeta}{1-\zeta} + \frac{-\rho}{1-\zeta}} \\ &= p_{vjd}(\omega_o)^{\frac{-1}{1-\zeta}} (P_{vjd})^{\frac{\rho-\zeta}{(1-\zeta)(\rho-1)}} \tilde{\lambda}_{vjd} (P_{jd})^{\frac{\rho}{1-\rho}} \eta_{jd} I_d \end{aligned}$$

Using the definitions of the substitution parameters:

$$\frac{-1}{1-\zeta} = -\xi$$

$$\frac{\rho-\zeta}{(\rho-1)(1-\zeta)} = \xi - \sigma$$

$$\frac{\rho}{1-\rho} = \sigma - 1$$

I arrive at the optimal expenditure equation (3).

Firms Firms hire a_{oj} workers to produce one unit of good j . They draw their differentiation parameter δ_{vj} from the distribution $\mathcal{D}$. To differentiate their variety, they incur additional variable costs: To cater to a specific taste in a destination, firms hire κ_{vjd} workers (**design-to-market**),

$$\frac{\partial \kappa_{vjd}}{\partial \delta_{vj}} > 0.$$

Differentiation costs further depend on **input endowments** ι_{ovj} in the firm's location o . For instance, more high-skill factor endowments can lower the cost of higher δ varieties.

$$\frac{\partial \iota_{ovj}}{\partial \delta_{vj}} > 0, \quad \frac{\partial^2 \iota_{ovj}}{\partial \delta_{vj} \partial HS_o} < 0.$$

In the baseline model, $\kappa_{vjd} = \kappa_{vj} \forall d$ and $\iota_{ovj} = 1 \forall o$. Firms maximize profits

$$\pi_{vj}(\omega_o) = \sum_{d=1}^K c_{vjd}(\omega_o)(p_{vjd}(\omega_o) - \tau_{od}\kappa_{vjd}\iota_{ovj}a_{oj}w_o) - w_of_j \quad (25)$$

by charging the unit price

$$p_{vjd}(\omega_o) = \frac{\xi}{\xi - 1} \tau_{od}\kappa_{vjd}\iota_{ovj}a_{oj}w_o = p_{ovjd}. \quad (26)$$

From (26) it is clear that allowing for designing-to-market is isomorphic to allowing firms to charge a destination-specific mark-up (**pricing-to-market**).

Equilibrium Given firms' pricing decision, all firms in the differentiated goods sector in country o make equal profits, $x_{vjd}(\omega_o) = x_{ovjd} \forall \omega_o \in \Omega_{ovj}$. The price index of a variety vj in destination d , P_{vjd} , becomes

$$P_{vjd} = \frac{\xi}{\xi - 1} \kappa_{vjd} \left[\sum_{o=1}^K (\iota_{ovj}a_{oj}w_o\tau_{od})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}} \quad (27)$$

where $M_{ovj} = \int_{\omega \in \Omega_{ovj}} d\omega$ is the number of active firms producing variety δ_{vj} in o , and $\sum_{v=1}^{V_j} M_{ovj} = M_{oj}$. Without destination-specific fixed costs, active firms sell everywhere.

Define country d 's *supplier access* for a variety v by

$$\widetilde{M}_{vjd} \equiv \sum_{o=1}^K \frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} (\iota_{ovj})^{1-\xi} M_{ovj}.$$

Note that supplier access now also depends on exporter-specific differentiation costs. Substituting P_{vjd} from (27) into the expression for P_{jd} in (4), using $\lambda_{vjd} = I_d^{\delta_{vj}}$, and rearranging yields

$$P_{jd}^{1-\sigma} = \left(\sum_{v=1}^{V_j} \alpha_{vjd} I_d^{\delta_{vj}} \left(\frac{\xi}{\xi-1} \kappa_{vjd} \left[\sum_{o=1}^K (\iota_{ovj} a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}} \right)^{1-\sigma} \right)$$

$$P_{jd}^{1-\sigma} k_{jd} = \left(\sum_{v=1}^{V_j} \alpha_{vjd} I_d^{\delta_{vj}} (\kappa_{vjd})^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}} \right) \quad (28)$$

where $k_{jd} \equiv \left(\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi} \right)^{\frac{1-\sigma}{\xi-1}} \left(\frac{\xi}{\xi-1} \right)^{\sigma-1}$.

Finally, substituting all price expressions in the demand equation (3), expenditure per firm equals

$$x_{ovjd} = \left(\frac{p_{ovjd}}{P_{vjd}} \right)^{1-\xi} \left(\frac{P_{vjd}}{P_{jd}} \right)^{1-\sigma} \alpha_{vjd} \lambda_{vjd} \eta_{jd} I_d$$

$$= \left(\frac{\frac{\xi}{\xi-1} \tau_{od} \kappa_{vjd} \iota_{ovj} a_{oj} w_o}{\frac{\xi}{\xi-1} \kappa_{vjd} \left[\sum_{o=1}^K (\iota_{ovj} a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}}} \right)^{1-\xi}$$

$$\frac{\left(\frac{\xi}{\xi-1} \kappa_{vjd} \left[\sum_{o=1}^K (\iota_{ovj} a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}} \right)^{1-\sigma}}{\sum_{v=1}^{V_j} \alpha_{vjd} I_d^{\delta_{vj}} \left(\frac{\xi}{\xi-1} \right)^{1-\sigma} (\kappa_{vjd})^{1-\sigma} \left(\left[\sum_{o=1}^K (a_{oj} w_o \tau_{od})^{1-\xi} (\iota_{ovj})^{1-\xi} M_{ovj} \right]^{\frac{1}{1-\xi}} \right)^{1-\sigma}} \alpha_{vjd} \lambda_{vjd} \eta_{jd} I_d$$

Collecting terms, we can write firm-level expenditure as

$$\begin{aligned}
x_{ovjd} &= \frac{1}{M_{ovj(d)}} s_{ovjd} s_{vjd} \eta_{jd} I_d \tag{29} \\
\text{where } s_{ovjd} &= \left(\frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} (\iota_{ovj})^{1-\xi} M_{ovj} \right) / \widetilde{M}_{vjd} \\
\text{and } s_{vjd} &= \frac{\alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vjd}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} I_d^{\delta_{v'j}} \kappa_{v'jd}^{1-\sigma} [\widetilde{M}_{v'jd}]^{\frac{1-\sigma}{1-\xi}}}
\end{aligned}$$

in equilibrium. The zero profit condition, which pins down the number of firms M_{ovj} per variety in equilibrium, is given by

$$\begin{aligned}
\Delta_{ovj} &\equiv \sum_{d=1}^K \frac{1}{M_{ovj}} \left(\frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} (\iota_{ovj})^{1-\xi} M_{ovj} \right) / \widetilde{M}_{vjd} \frac{\alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vjd}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} I_d^{\delta_{v'j}} \kappa_{v'jd}^{1-\sigma} [\widetilde{M}_{v'jd}]^{\frac{1-\sigma}{1-\xi}}} \eta_{jd} I_d \\
&\Leftrightarrow \iota_{ovj}^{1-\xi} \sum_{d=1}^K \left(\frac{(\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} \right) \frac{\alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vjd}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} I_d^{\delta_{v'j}} \kappa_{v'jd}^{1-\sigma} [\widetilde{M}_{v'jd}]^{\frac{1-\sigma}{1-\xi}}} \eta_{jd} I_d \leq \frac{\xi w_o f_j}{(a_{oj} w_o)^{1-\xi}}. \\
&\Leftrightarrow \Delta_{ovj} \equiv \iota_{ovj}^{1-\xi} \sum_{d=1}^K (\tau_{od})^{1-\xi} \frac{s_{vjd} \eta_{jd} I_d}{\left[\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd} \iota_{o'vj})^{1-\xi} M_{o'vj} \right]} \leq \frac{\xi w_o f_j}{(a_{oj} w_o)^{1-\xi}}. \tag{30}
\end{aligned}$$

Similar to the baseline, firms' entry decision hinges on market access. In addition, entry depends on the exporter-specific differentiation costs. Higher quality goods are more costly to produce. In countries with more high-skilled workers, however, differentiation is relatively cheaper. Thus, higher skill endowment mutes the entry deterring effect of the differentiation cost.

A.2 Analytical results

Proof of Proposition 1 The income elasticity of demand for goods from a given origin is

$$\varepsilon_{ojd} \equiv \frac{dX_{ojd}}{dI_d} \frac{I_d}{X_{ojd}} = 1 + \sum_{v=1}^{V_j} \frac{s_{ovjd} s_{vjd}}{\sum_{v'=1}^{V_j} s_{ov'jd} s_{v'jd}} \delta_{vj} - \sum_{v=1}^{V_j} s_{vjd} \delta_{vj}. \tag{31}$$

For $\varepsilon_{ojd} > 1$:

$$\begin{aligned} & \sum_{v=1}^{V_j} \frac{s_{ovjd} s_{vjd}}{\sum_{v'=1}^{V_j} s_{ov'jd} s_{v'jd}} \delta_{vj} - \sum_{v'=1}^{V_j} s_{v'jd} \delta_{v'j} > 0 \\ \Leftrightarrow & \sum_{v=1}^{V_j} s_{vjd} s_{ovjd} \delta_{vj} - \left(\sum_{v'=1}^{V_j} s_{v'jd} \delta_{v'j} \right) \left(\sum_{v'=1}^{V_j} s_{v'jd} s_{ov'jd} \right) > 0. \end{aligned} \quad (32)$$

Substituting the equilibrium expression for s_{ovjd} from equation (29) gives

$$\begin{aligned} & \sum_{v=1}^{V_j} s_{vjd} \theta_{ojd} \frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}} \delta_{vj} - \left(\sum_{v'=1}^{V_j} s_{v'jd} \delta_{v'j} \right) \left(\sum_{v'=1}^{V_j} s_{v'jd} \theta_{ojd} \frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}} \right) > 0 \\ \Leftrightarrow & \sum_{v=1}^{V_j} s_{vjd} \frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}} \delta_{vj} - \left(\sum_{v'=1}^{V_j} s_{v'jd} \delta_{v'j} \right) \left(\sum_{v'=1}^{V_j} s_{v'jd} \frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}} \right) > 0 \end{aligned} \quad (33)$$

Based on Anderson et al. (1987) and Bohr et al. (2023), since $s_{vjd} \geq 0$ and $\sum_v s_{vjd} = 1$, we can re-interpret the destination variety expenditure shares as the probability of destination d to consume variety vj . Treating s_{vjd} as such probability measure, i.e. $\mathbb{E}_s[\phi_v] \equiv \sum_v s_{vjd} \phi_v$, we can re-write the condition for $\varepsilon_{ojd} > 1$ as

$$Cov_{ojd} \left(\frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}}, \delta_{vj} \right) > 0, \quad (34)$$

where the index reflects that the covariance varies across country pairs and products because the probability measure varies at the destination-product level and the number of firms varies at the origin level.

Proposition 2 Up to first order, the covariance can be approximated by

$$\begin{aligned} & Cov_{ojd} \left(\frac{(\iota_{ovj})^{1-\xi} M_{ovj}}{\widetilde{M}_{vjd}}, \delta_{vj} \right) \approx Var_{jd}(\delta_{vj}) \\ & \times \left[\mathbb{E} \left(\frac{M_{ovj(d)}}{\widetilde{M}_{vjd}} \underbrace{\frac{\partial (\iota_{ovj})^{1-\xi}}{\partial \delta_{vj}}}_{<0} \right) + \mathbb{E} \left(\frac{(\iota_{ovj})^{1-\xi}}{\widetilde{M}_{vjd}} \underbrace{\frac{\partial M_{ovj}}{\partial \delta_{vj}}}_{\text{IFT threshold}} \right) - \mathbb{E} \left(\frac{(\iota_{ovj})^{1-\xi} M_{ovj(d)}}{\widetilde{M}_{vjd}^2} \underbrace{\frac{\partial \widetilde{M}_{vjd}}{\partial \delta_{vj}}}_{<0} \right) \right] \end{aligned} \quad (35)$$

Relative to the baseline model, two additional forces determine the sign of the covariance: First, exporter-specific differentiation costs deter entry in high-end varieties.

This drives as well the second additional effect, namely that supplier access tends to be weaker for high-end varieties. Important for the empirical strategy is that the supplier access-effect is uniform across exporters and accounted for by destination-product fixed effects. Disproportionate exports to high-income countries can therefore still be attributed to the distribution of firms across quality levels in the exporting country, conditional on average differentiation costs.

The supplier access-effect is given by

$$\frac{\partial \widetilde{M}_{vjd}}{\partial \delta_{vj}} = (1 - \xi) \sum_{o=1}^K \frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} (\iota_{ovj})^{1-\xi} M_{ovj(d)} \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{1}{\iota_{ovj}} < 0 \quad (36)$$

By the implicit function theorem, the firm distribution across quality levels is determined by

$$\frac{\partial M_{ovj}}{\partial \delta_{vj}} = - \frac{\partial \Delta_{ovj} / \partial \delta_{vj}}{\partial \Delta_{ovj} / \partial M_{ovj}}, \quad (37)$$

where

$$\Delta_{ovj} \equiv \iota_{ovj}^{1-\xi} \sum_{d=1}^K (\tau_{od})^{1-\xi} \frac{s_{vjd} \eta_{jd} I_d}{\left[\sum_{o=1}^K (a_{oj} w_o \tau_{od} \iota_{ovj})^{1-\xi} M_{ovj} \right]}.$$

and define

$$\check{M}_{vjd} \equiv \sum_{o=1}^K (a_{oj} w_o \tau_{od} \iota_{ovj})^{1-\xi} M_{ovj}.$$

Using

$$\frac{\partial s_{vjd}}{\partial M_{ovj}} = (1 - s_{vjd}) \frac{1 - \sigma}{1 - \xi} \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial M_{ovj}} \quad (38)$$

and

$$\frac{\partial \check{M}_{vjd}}{\partial M_{ovj}} = (a_{oj} w_o \tau_{od} \iota_{ovj})^{1-\xi} \quad (39)$$

yields

$$\begin{aligned}
\frac{\partial \Delta_{ovj}}{\partial M_{ovj}} &= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \left[\frac{\partial s_{vjd} / \partial M_{ovj}}{\check{M}_{vjd}} - \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial M_{ovj}} \frac{1}{\check{M}_{vjd}} \right] \\
&= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \left[\frac{1}{\check{M}_{vjd}} \left((1 - s_{vjd}) \frac{1 - \sigma}{1 - \xi} \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial M_{ovj}} \right) - \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial M_{ovj}} \frac{1}{\check{M}_{vjd}} \right] \\
&= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \left(\frac{1}{\check{M}_{vjd}} \frac{s_{vjd}}{\check{M}_{vjd}} \underbrace{\frac{\partial \check{M}_{vjd}}{\partial M_{ovj}}}_{>0} \right) \underbrace{\left[\left((1 - s_{vjd}) \frac{1 - \sigma}{\underbrace{1 - \xi}_{<1}} \right) - 1 \right]}_{<0} < 0.
\end{aligned} \tag{40}$$

Competition continues to unambiguously deter entry in the generalized model.

Next, using

$$\frac{\partial s_{vjd}}{\partial \delta_{vj}} = (1 - s_{vjd}) s_{vjd} \left[\ln(I_d) + (1 - \sigma) \frac{1}{\kappa_{vjd}} \frac{\partial \kappa_{vjd}}{\partial \delta_{vj}} + \frac{1 - \sigma}{1 - \xi} \frac{1}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} \right] \tag{41}$$

$$\frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} = \sum_{o=1}^K (a_{oj} w_o \tau_{od} \iota_{ovj})^{1-\xi} M_{ovj} \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{1}{\iota_{ovj}} (1 - \xi) \tag{42}$$

yields

$$\begin{aligned}
\frac{\partial \Delta_{ovj}}{\partial \delta_{vj}} &= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \left[\frac{\partial s_{vjd} / \partial \delta_{vj}}{\check{M}_{vjd}} - \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} \frac{1}{\check{M}_{vjd}} \right] + (1-\xi) \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{\Delta_{ovj}}{\iota_{ovj}} \\
&= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{1}{\check{M}_{vjd}} \left[\frac{\partial s_{vjd}}{\partial \delta_{vj}} - \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} \right] + (1-\xi) \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{\Delta_{ovj}}{\iota_{ovj}} \\
&= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{1}{\check{M}_{vjd}} \left[(1-s_{vjd}) s_{vjd} \left[\ln(I_d) + (1-\sigma) \frac{1}{\kappa_{vjd}} \frac{\partial \kappa_{vjd}}{\partial \delta_{vj}} \right] \right] \\
&\quad + \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{1}{\check{M}_{vjd}} \left[(1-s_{vjd}) s_{vjd} \left[\frac{1-\sigma}{1-\xi} \frac{1}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} \right] - \frac{s_{vjd}}{\check{M}_{vjd}} \frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}} \right] \\
&\quad + (1-\xi) \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{\Delta_{ovj}}{\iota_{ovj}} \\
&= \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} \left[(1-s_{vjd}) \left[\ln(I_d) + (1-\sigma) \frac{1}{\kappa_{vjd}} \frac{\partial \kappa_{vjd}}{\partial \delta_{vj}} \right] \right] \\
&\quad + \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} \frac{1}{\check{M}_{vjd}} \underbrace{\frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}}}_{<0} \underbrace{\left[(1-s_{vjd}) \left[\frac{1-\sigma}{1-\xi} \right] - 1 \right]}_{<0} + (1-\xi) \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{\Delta_{ovj}}{\iota_{ovj}}
\end{aligned} \tag{43}$$

Variation in market access across varieties implies a **threshold condition** for a positive association of the number of suppliers of a variety and its quality:

$$\begin{aligned}
&\sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} (1-s_{vjd}) \left[\ln(I_d) + \frac{1}{\check{M}_{vjd}} \underbrace{\frac{\partial \check{M}_{vjd}}{\partial \delta_{vj}}}_{<0} \underbrace{\left[\frac{1-\sigma}{1-\xi} - \frac{1}{1-s_{vjd}} \right]}_{<0} \right] \\
&> (\sigma-1) \sum_{d=1}^K (\tau_{od} \iota_{ovj})^{1-\xi} \eta_{jd} I_d \frac{s_{vjd}}{\check{M}_{vjd}} \left[(1-s_{vjd}) \frac{1}{\kappa_{vjd}} \frac{\partial \kappa_{vjd}}{\partial \delta_{vj}} \right] + (\xi-1) \frac{\partial \iota_{ovj}}{\partial \delta_{vj}} \frac{\Delta_{ovj}}{\iota_{ovj}}
\end{aligned} \tag{44}$$

In the generalized model, four factors play a role: First, proximity to demand for high-quality varieties induces entry. Second, lower competition in high-end varieties induces entry. Third and fourth, higher differentiation costs deter entry.

A.3 Sketch of Proof for Corollary 1.

If $MP_{ovj} > MP_{o'vj}$ and $\frac{\partial \Delta_{ovj}}{\partial M_{ovj}} = \frac{\partial \Delta_{o'vj}}{\partial M_{o'vj}}$, then $\frac{\partial M_{ovj}}{\partial \delta_{vj}} > \frac{\partial M_{o'vj}}{\partial \delta_{vj}}$, so that $Cov_{ojd} > Cov_{o'jd}$. Hence, $\varepsilon_{ojd} > \varepsilon_{o'jd}$.

B From theory to estimation

The derivations in this section are for the model given in the main text, section 2. In other words, $\kappa_{vjd} = \kappa_{vj}$ and $\iota_{ovj} = 1$ in the following.

B.1 First-order Taylor approximation

Bilateral trade per product is given by

$$X_{jod} = \sum_{v=1}^{V_j} M_{ovj} x_{oj}^\omega(\mathbf{B}_v, A_{ojd}), \quad (45)$$

where I summarize variety characteristics in the vector $\mathbf{B}_v = (\alpha_{vjd}, \delta_{vj}, M_{ovj})$ and product-exporter-destination characteristics in $A_{ojd} = (w_o, a_{oj}, \{\tau_{od}\}_o, I_d)$. Around an equilibrium without vertical product differentiation, i.e. when

$$\mathbf{B}_v = \mathbf{B} \equiv \left(\frac{1}{V_j} \sum_{v=1}^{V_j} \alpha_{vjd}, \frac{1}{V_j} \sum_{v=1}^{V_j} \delta_{vj}, \frac{1}{V_j} \sum_{v=1}^{V_j} M_{ovj} \right) = (\bar{\alpha}_{jd}, \bar{\delta}_j, \bar{M}_{oj}),$$

a first-order Taylor approximation of firm level exports equals

$$\begin{aligned} x_{oj}^\omega(\mathbf{B}_v, A_{ojd}) &\approx x_{oj}^\omega(\mathbf{B}, A_{ojd}) \\ &+ \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \alpha_{vjd}} \Big|_{\mathbf{B}_v=\mathbf{B}} (\alpha_{vjd} - \bar{\alpha}_{jd}) + \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \delta_{vj}} \Big|_{\mathbf{B}_v=\mathbf{B}} (\delta_{vj} - \bar{\delta}_j) \\ &+ \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial M_{ovj}} \Big|_{\mathbf{B}_v=\mathbf{B}} (M_{ovj} - \bar{M}_{oj}). \end{aligned} \quad (46)$$

V_j is the number of different varieties for product j . Given the model structure, we have:

$$x_{oj}^\omega(\mathbf{B}_v, A_{ojd}) = \frac{1}{M_{ovj}} \underbrace{\left(\frac{(a_{oj} w_o \tau_{od})^{1-\xi} M_{ovj}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi} M_{o'vj}} \right)}_{=s_{ovjd}} \underbrace{\frac{\alpha_{vjd} I_d^{\delta_{vj}} \kappa_{vj}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}{\sum_{v'=1}^{V_j} \alpha_{v'jd} I_d^{\delta_{v'j}} \kappa_{v'j}^{1-\sigma} [\widetilde{M}_{vjd}]^{\frac{1-\sigma}{1-\xi}}}}_{=s_{vjd}} \eta_{jd} I_d$$

which, evaluated at $\mathbf{B}$, becomes

$$x_{oj}^\omega(\mathbf{B}, A_{ojd}) = \frac{1}{\bar{M}_{oj}} \underbrace{\left(\frac{(a_{oj}w_o\tau_{od})^{1-\xi} \bar{M}_{oj}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi} \bar{M}_{o'j}} \right)}_{=\bar{s}_{ojd}} \underbrace{\frac{1}{V_j}}_{=\bar{s}_j} \eta_{jd} I_d. \quad (47)$$

Next,

$$\begin{aligned} \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \alpha_{vjd}} &= x_{oj}^\omega(\cdot) \frac{1}{\alpha_{vjd}} (1 - s_{vjd}) \\ \left. \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \alpha_{vjd}} \right|_{\mathbf{B}_v=\mathbf{B}} &= x_{oj}^\omega(\cdot) \frac{1}{\bar{\alpha}_{jd}} \left(1 - \frac{1}{V_j} \right) \end{aligned} \quad (48)$$

$$\frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \delta_{vj}} = x_{oj}^\omega(\cdot) \ln(I_d) (1 - s_{vjd}) + x_{oj}^\omega(\cdot) \frac{1}{\kappa_{vj}} (1 - \sigma) \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} (1 - s_{vjd})$$

$$\left. \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial \delta_{vj}} \right|_{\mathbf{B}_v=\mathbf{B}} = x_{oj}^\omega(\cdot) \ln(I_d) \left(1 - \frac{1}{V_j} \right) + x_{oj}^\omega(\cdot) \frac{1}{\kappa_{vj}} (1 - \sigma) \left. \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} \right|_{\delta_{vj}=\bar{\delta}_j} \left(1 - \frac{1}{V_j} \right) \quad (49)$$

$$\begin{aligned} \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial M_{ovj}} &= x_{oj}^\omega(\cdot) \left(- \left(\frac{(a_{oj}w_o\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi} \bar{M}_{o'vj}} \right) + \beta \frac{1}{\bar{M}_{vjd}} \frac{(a_{oj}w_o\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} (1 - s_{vjd}) \right) \\ &= x_{oj}^\omega(\cdot) \left(\beta \frac{1 - s_{vjd}}{\bar{M}_{vjd}} \frac{(a_{oj}w_o\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} - \frac{(a_{oj}w_o\tau_{od})^{1-\xi} / \sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}}{\sum_{o'=1}^K \frac{(a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} \bar{M}_{o'vj}} \right) \\ &= x_{oj}^\omega(\cdot) \left(\frac{1}{\bar{M}_{vjd}} \frac{(a_{oj}w_o\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} (\beta(1 - s_{vjd}) - 1) \right), \end{aligned}$$

where $\beta = \frac{1-\sigma}{1-\xi}$ (for readability).

$$\left. \frac{\partial x_{oj}^\omega(\mathbf{B}_v, A_{ojd})}{\partial M_{ovj}} \right|_{\mathbf{B}_v=\mathbf{B}} = x_{oj}^\omega(\cdot) \left(\frac{1}{\bar{M}_{jd}} \frac{(a_{oj}w_o\tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} \right) \left(\beta(1 - \frac{1}{V_j}) - 1 \right) \quad (50)$$

Substituting all terms, equation (46) becomes

$$\begin{aligned}
x_{oj}^\omega(\mathbf{B}_v, A_{ojd}) &\approx x_{oj}^\omega(\cdot) \left[1 + \left(1 - \frac{1}{V_j} \right) \frac{1}{\bar{\alpha}_{jd}} (\alpha_{vjd} - \bar{\alpha}_{jd}) \right. \\
&\quad + \left(1 - \frac{1}{V_j} \right) \left(\ln(I_d) + \frac{1}{\kappa_{vj}} (1 - \sigma) \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} \Big|_{\delta_{vj}=\bar{\delta}_j} \right) (\delta_{vj} - \bar{\delta}_j) \\
&\quad \left. + \left(\frac{1 - \sigma}{1 - \xi} \left(1 - \frac{1}{V_j} \right) - 1 \right) \left(\frac{1}{\bar{M}_{jd}} \frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} \right) (M_{ovj} - \bar{M}_{oj}) \right].
\end{aligned} \tag{51}$$

Now note that for any random variable $x_{vj} = \{\alpha_{vjd}, \delta_{vj}, M_{ovj}\}$

$$\begin{aligned}
\sum_{v=1}^{V_j} M_{ovj} (x_{vj} - \bar{x}_j) &= \left(\sum_{v=1}^{V_j} M_{ovj} x_{vj} \right) - M_{oj} \bar{x}_j \\
&= M_{oj} \left[\frac{1}{\bar{M}_{oj}} \left(\sum_{v=1}^{V_j} M_{ovj} x_{vj} \right) - \bar{x}_j \right] \\
&= M_{oj} \left[\frac{1}{\bar{M}_{oj}} \left(\sum_{v=1}^{V_j} M_{ovj} x_{vj} \right) - \bar{x}_j \bar{M}_{oj} \frac{1}{\bar{M}_{oj}} \right] \\
&= M_{oj} \frac{1}{\bar{M}_{oj}} \left[\left(\frac{1}{V_j} \sum_{v=1}^{V_j} M_{ovj} x_{vj} \right) - \bar{x}_j \bar{M}_{oj} \right] \\
&= M_{oj} \frac{Cov_{oj}(M_{ovj}, x_{vj})}{\bar{M}_{oj}} \\
&= V_j Cov_{oj}(M_{ovj}, x_{vj})
\end{aligned} \tag{52}$$

where the third equality uses that $M_{oj} = V_j \frac{1}{V_j} \sum_{v=1}^{V_j} M_{ovj} = V_j \bar{M}_{oj}$. Of course, when $x_{vj} = M_{ovj}$, then $Cov_{oj}(M_{ovj}, x_{vj}) = Var_{oj}(M_{ovj})$.

Taking logs of (51), using $\log(1+x) \approx x$ when x is small, aggregating across varieties, and using (52) yields

$$\begin{aligned}
\log(X_{jod}) &\approx M_{oj} \log(x_{oj}^\omega(\mathbf{B}, A_{ojd})) \\
&\quad + (V_j - 1) \left(\ln(I_d) + \frac{1}{\kappa_j} (1 - \sigma) \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} \Big|_{\delta_{vj}=\bar{\delta}_j} \right) Cov_{jo}(M_{ovj}, \delta_{vj}) \\
&\quad + \left(\frac{1 - \sigma}{1 - \xi} (V_j - 1) - V_j \right) \left(\frac{(a_{oj} w_o \tau_{od})^{1-\xi}}{\sum_{o'=1}^K (a_{o'j} w_{o'} \tau_{o'd})^{1-\xi}} \bar{M}_{oj} \right) Var_{jo}(M_{ovj}) \\
&\quad + (V_j - 1) \frac{1}{\bar{\alpha}_{jd}} Cov_{jo}(M_{ovj}, \alpha_{vjd}).
\end{aligned} \tag{53}$$

The first order Taylor approximation allows to write the gravity equation into a homothetic part and a part related to the decomposition of the variety-specific price index P_{vjd} .

The first term in (53) is (similar to) a homothetic gravity equation. The remaining three terms introduce a “correction” for the product differentiation. The coefficient $(V_j - 1)$ shows that the importance of the correction varies across products. It is more important the more varieties a product has, i.e. the higher the scope for product differentiation. The last term in each of the three correction terms shows that the correction matters more for exporters with a more skewed firm distribution (size distribution of variety v), i.e. when the firm distribution deviates more from a uniform distribution ($\frac{M_{ovj}}{M_{oj}}$ vs $1/V_j$).

The second term in (53) relates to the non-homotheticity in demand. Two effects are present: The destination income raises demand for goods from countries with more firms in the high-end distribution. However, high-end varieties are costly to produce ($\frac{\partial \kappa_{vj}}{\partial \delta_{vj}} > 0$). This compresses demand for goods from high-end manufacturing countries. The cost component only varies by exporter-product and thus will be absorbed in the product-exporter-time fixed effect.

The third term captures the supplier access component of both price indices. The coefficient is negative since $\frac{1-\sigma}{1-\xi} < 1$. Since trade costs enter with a negative exponent, the term becomes smaller for trade between far-away countries and is negligible for the estimation. Trade between nearby-countries is attenuated more if (i) the dispersion of firms across varieties in the origin is high - without dispersion, the product level price index is only determined by differentiation costs and taste (trade costs drop out); (ii) marginal production costs in the origin are low - shifting higher weight to trade costs as a price determinant for goods from o ; and (iii) supplier access in the destination is low - raising the price index for a variety.

The fourth term in (53) can be interpreted as a structural error term as it only relates to the taste component α_{vjd} .

Finally, equation (16) is given by

$$\ln(X_{jod}) \approx \psi_{jo} + \varphi_{jd} + f_{jod}(\tau_{od}) + \beta_j \ln(I_d) Cov_{jo}(M_{ovj}, \delta_{vj}) + \nu_{jod} \quad (54)$$

where

$$\begin{aligned} \psi_{jo} &= M_{jo}(1 - \xi_j) \ln(a_{oj}w_o) + (V_j - 1)(1 - \sigma) \frac{1}{\kappa_j} \frac{\partial \kappa_{vj}}{\partial \delta_{vj}} \Big|_{\delta_{vj}=\bar{\delta}_j} Cov_{jo}(M_{ovj}, \delta_{vj}) \\ \varphi_{jd} &= \ln \left(\frac{1}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi}} \frac{1}{\bar{M}_{o'j}} \frac{1}{V_j} \eta_{jd} I_d \right) \\ \beta_j &= (V_j - 1) \\ \nu_{jod} &= (V_j - 1) \frac{1}{\bar{\alpha}_{jd}} Cov_{jo}(M_{ovj}, \alpha_{vjd}) \end{aligned}$$

and

$$f_{jod}(\tau_{od}) = \gamma_j^1 \ln(\tau_{od}) + \gamma_{ojd}(\tau_{od}) \gamma_j^1 \quad (55)$$

$$\begin{aligned} \gamma_j^1 &= (1 - \xi_j) \\ \gamma_{ojd} &= \left(\frac{1 - \sigma}{1 - \xi} (V_j - 1) - V_j \right) Var_{oj}(M_{ovj}) \left(\frac{(a_{oj}w_o)^{1-\xi}}{\sum_{o'=1}^K (a_{o'j}w_{o'}\tau_{o'd})^{1-\xi} \bar{M}_{o'j}} \right). \end{aligned}$$

under the assumption that $\gamma_{ojd}(\tau_{od}) \gamma_j^1 \rightarrow 0$ for observed values of τ_{od} so that $f_{jod}(\tau_{od}) \approx \gamma_j^1 \ln(\tau_{od})$.

C Data

Table 7 provides summary statistics of the key dimensions of the sample. The effect of per capita income interacted with market potential is identified from variation across on average 113 destinations and 79 exporters. For each importer-time-HS 4 triad, there are at least 6 exporters. For each exporter-time-HS 4 triad, the sample includes at least 14 importer.

Table 7: Summary statistics of regression sample

	Mean	Sd	Median	Min	Max
Importer-time per exporter & time	112	36	126	15	150
Exporter-time per importer & time	107	28	113	15	146
Exporter per importer & time & hs4	36	18	32	15	138
Importer per exporter & time & hs4	61	33	53	15	150

Notes: The table presents summary statistics on the number of observations along key dimensions in the trade data.

Table 8: Summary statistics on the unit value variance across destination-years, by exporter-hs4

	Mean	Sd	P25	Median	P75	N
Std. dev. of residualized unit values, by hs4-exp	1.13	0.37	0.87	1.08	1.33	31,806
Weighted mean sd(uv), by hs4	0.84	0.26	0.68	0.79	0.92	436
Weighted mean sd(uv), by exp.	0.96	0.25	0.80	0.94	1.06	149

Notes:

D Robustness checks

First, I replicate the baseline regressions using the Pseudo Poisson Maximum Likelihood (PPML) estimator. Since Silva and Tenreyro (2006), the PPML estimator has become a standard estimation method for gravity regressions because PPML can accommodate heteroscedasticity in the trade data.¹⁷ Figure 7 plots the PPML estimates against the baseline OLS estimates (whether or not based on a sample including zero

¹⁷However, to the best of my knowledge, all studies on the topic of quality in trade use OLS estimation.

trade flows). Aside from some differences in significance - I plot insignificant coefficients as zeros - the results are consistent across specifications. I therefore continue with OLS estimation for the remainder of the paper.

Second, I control for exporter-destination fixed effects instead of relying on time-invariant proxies to measure trade costs. This strategy is motivated by the recommendations for gravity estimation by Yotov et al. (2016) and Larch et al. (2025). In this specification, the coefficients of interest on the interaction between destination per capita income and market potential bins are identified from time variation in the two variables. Results are presented in Table 9. In this very restrictive setup, the non-linearity is still clearly present, albeit less than a quarter of all products included in the regression show significant effects for the interaction term. The Wald test of joint nullity of the bin-coefficients is rejected for roughly 200 products.

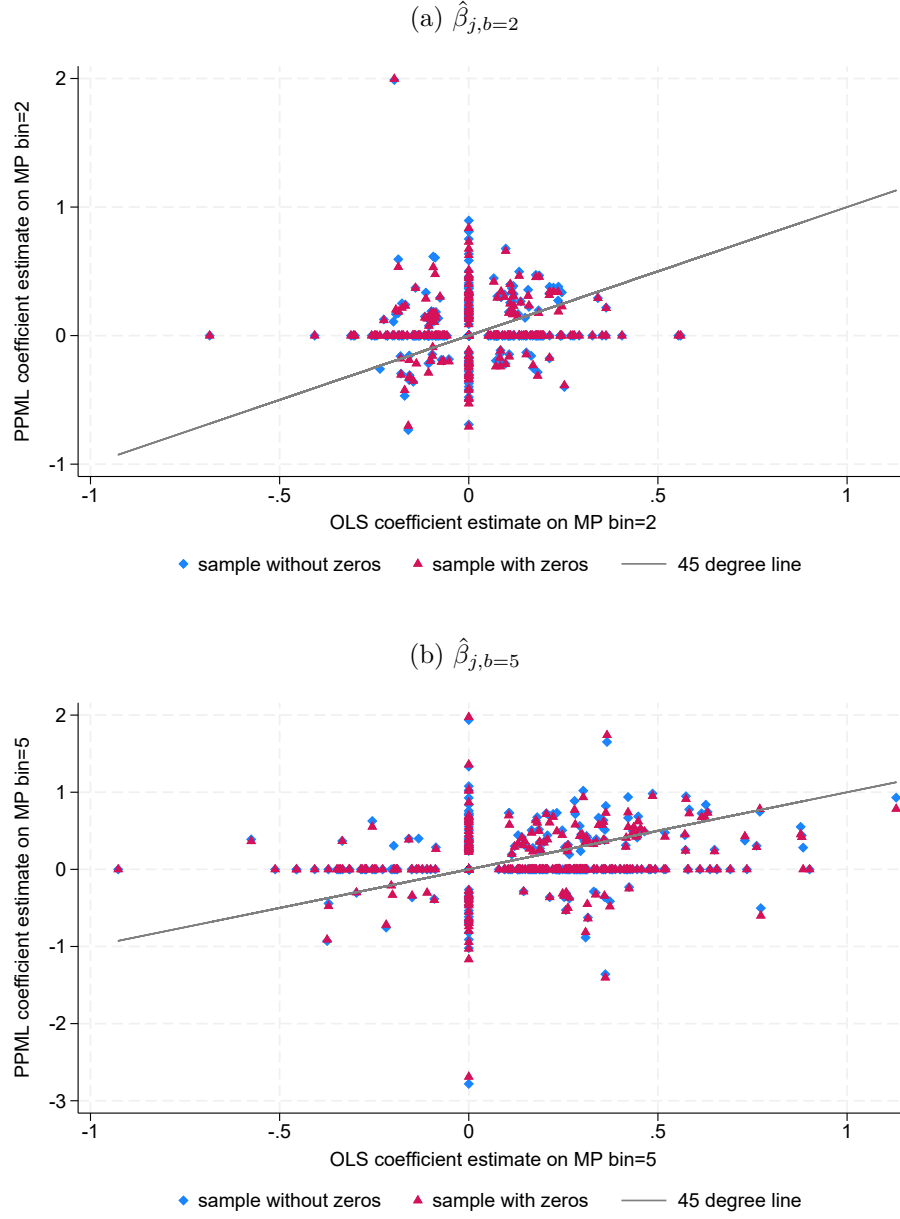
Table 9: Summary statistics for coefficient estimates, robustness with *od* fixed effects

	Mean	SD	p25	p50	p75	# HS4 sig.
FMP bin 2	-0.03	0.15	-0.11	-0.06	0.07	69
FMP bin 3	-0.05	0.13	-0.13	-0.08	0.07	80
FMP bin 4	0.00	0.20	-0.13	0.07	0.13	77
FMP bin 5	0.10	0.23	0.10	0.14	0.21	87
$\log(y_o)$	-0.01	0.11	-0.07	-0.04	0.05	115
RTA	-0.04	0.29	-0.19	-0.14	0.14	73
WTO	-0.58	0.98	-0.97	-0.61	-0.43	63
Number of HS4 codes	431					

Notes: Summary statistics for parameter estimates. HS4-level regressions control for exporter-year, importer-year and exporter-importer fixed effects. The last column shows the number of products for which the parameter estimates are significantly different from 0 at the 5% confidence level. We cannot reject the null hypothesis of joint nullity of market potential bins for 236 products. The total number of products is 431.

Third, I vary the clustering of the standard errors. Table 10 presents the results in terms of the share of products for which each market potential coefficient is significant. The first row reports the statistics for the baseline estimation. The second row reports the shares when standard errors are clustered to allow correlations across destination-years within exporter-product pairs and correlations across exporter-years within destination-product pairs. The share of products with significant estimates drops across quintiles yet is still highest for the top market potential quintile. Thus, the non-linearity result holds even with this more lenient way of treating standard

Figure 7: Correlation of OLS and PPML estimates



Notes: Panel (a) shows the correlation of the coefficient estimates of $\beta_{j,b=2}$ from estimation of (19) when estimated using OLS (baseline) or PPML. Panel (b) shows the correlation of the coefficient estimates of $\beta_{j,b=5}$ from estimation of (19) when estimated using OLS (baseline) or PPML. Whether or not the sample includes zero trade flows is indicated in the plot legend. Zeros are defined as trade flows between countries that trade a given product in a given year but not with each other. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for calculations.

error correlations.

Table 10: Share of products with statistically significant coefficient by cluster variable(s)

	FMP bin 2	FMP bin 3	FMP bin 4	FMP bin 5
<i>jod</i>	0.44	0.50	0.49	0.57
<i>jo, jd</i>	0.18	0.18	0.15	0.25

Notes: Table shows share of products with coefficients statistically different from 0 at the 5% significance level in the baseline regression model when standard errors are clustered by country pair (row 1) or by exporter and importer separately (row 2).

The fourth robustness check pertains to the construction of market potential. Results shown in Table 11 confirm the baseline result for different ways of constructing market potential. The table reports rank correlation coefficients of the baseline estimates with estimates of the baseline regression based on different empirical market potential measures. Overall, the correlation among the coefficients on the top market potential quintile tends to be higher than the correlation of coefficients of the second market potential quintile, with the other two coefficient groups somewhere in the middle. To start, I vary the set of countries included in the construction of market potential. Consistent with theory, market potential in row (1) includes the domestic per capita income. Market potential in row (2) is defined over the set of trade partners only while in row (3) it is defined over the set of neighboring countries only. The main results appear to hold qualitatively for all three alternative market potential definitions. The weakest correlation is with the specification of considering only neighboring countries.

Next, I vary the measurement of per capita income entering the weighted average. First, I correct per capita income for within-country inequality where data on inequality is available. I report the results in the middle panel of Table 11. For comparability, I re-construct the baseline market potential with only those countries for which inequality data is available in the WIID database. To correct for inequality, I regress log GDP per capita on the Gini coefficient and its square and compute market potential as the weighted average of the residual of this regression instead of the raw per capita income. Thereby, I leverage variation in per capita income beyond what can be explained with within-country inequality. The correlation coefficient between estimates on the inequality-corrected market potential and its benchmark are high. Second, I

fix per capita income to its initial level so that identification of the coefficients is fully due to cross-section variation as in the model. The correlation with the baseline result remains high as well.¹⁸

Lastly, I change the weights to construct market potential. The bottom two rows in Table 11 report the correlation for these robustness checks. The first row of that panel reports results when income per capita is weighted by the product of population and inverse distance to the market, inspired by Dingel (2017). Market potential does not vary across products in this case. The bottom row shows the results when the weights are based on estimates of equation (18) using the ITPD-E database. While the database measures trade at the industry-level (more aggregate than HS 2-digit), its advantage is that it covers domestic trade flows. The inclusion of domestic trade flows accounts for competition from domestic suppliers in the estimation of the destination-product market scaling factor F_{jmt} and thus improves the estimate (though at a more aggregate level). Changing the weights in either way results in similar deviations from the baseline. The correlations are lowest among the robustness checks pointing to the importance of accounting for the disaggregated weights in line with theory.

Table 11: Robustness to market potential measurement

		FMP bin 2	FMP bin 3	FMP bin 4	FMP bin 5
change sample	incl. home	0.69	0.78	0.76	0.76
	only trade partners	0.48	0.55	0.62	0.72
	only neighbors	0.20	0.25	0.36	0.40
change y_{mt}	correct for inequality	0.56	0.67	0.66	0.70
	fix y_{mt} at $t = 0$	0.37	0.48	0.57	0.76
change weights	inverse distance x pop.	0.15	0.21	0.33	0.39
	include border effect	0.03	0.11	0.24	0.28

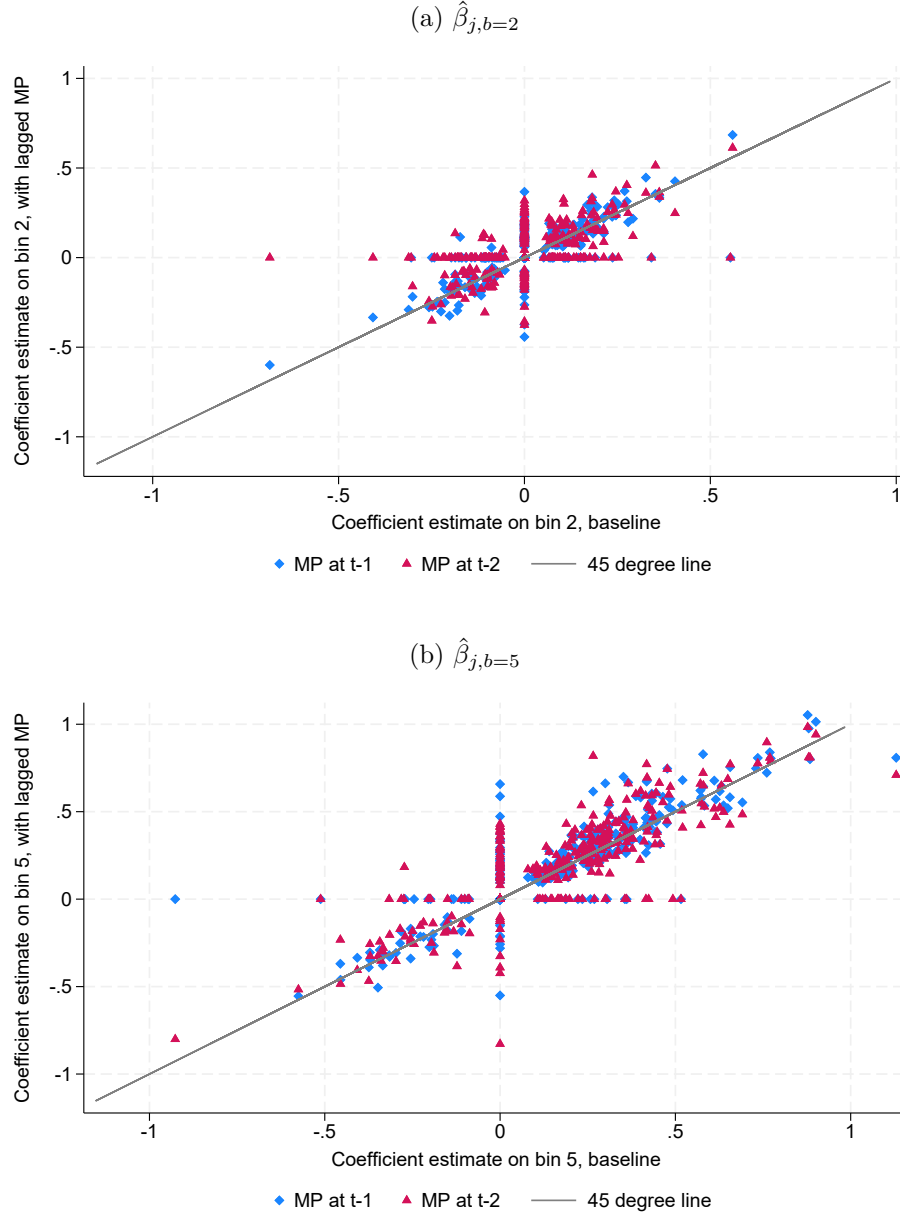
Notes: Spearman correlation coefficients with baseline estimates. Row 4 reports the Spearman correlation with the baseline specification, but fixing the sample to the one for the robustness exercise. See text for details. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0.

As a fifth robustness check, I explore the dynamics of the effect. I replace market potential and exporter per capita income with their one- or two-period lags. The results are presented in Figure 8. The most notable difference is in the significance. Insignificant coefficients are set to zero but still plotted. Aside from the deviations in terms of significance, approximately two-thirds of coefficients on the top quintile

¹⁸Variation over time is solely driven by entry of regional trade agreements or WTO accession which is rare.

dummy are higher when estimated on the lagged version of market potential than in the baseline. This could speak to a time lag between firms observing their market potential and transforming it in market access, i.e. actual sales. From an econometric viewpoint, trade could reversely affect the *exporters'* contemporaneous per capita income and their market potential. In expectation this would induce an *upward* bias in the baseline regression as one could expect trade with richer countries to be beneficial for exporter growth. The results in the plot, if anything, go in the opposite direction, however. Given the disaggregate level of the analysis and the rich set of fixed effects, this reverse causality channel seems not to be at play. In section 5.2, I discuss reverse causality induced by the effect of trade on *destination* per capita income.

Figure 8: Correlation of contemporaneous and lagged market potential effects



Notes: Panel (a) shows the correlation of the coefficient estimates of $\beta_{j,b=2}$ from estimation of (19) when estimated using the baseline or a lagged version of market potential. Panel (b) shows the correlation of the coefficient estimates of $\beta_{j,b=5}$ from estimation of (19) when estimated using the baseline or a lagged version of market potential. Coefficient estimates that are not statistically different from zero at the 5% significance level are set to 0 for calculations.